

From cells to councils: a conservation law of allocation under scarcity

James Miller

Independent researcher

contact@n-ought.com

ORCID: 0009-0009-6595-647X

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Abstract

Under scarcity, allocation is zero-sum. In a system short of a resource, no ordering of access reduces the total shortfall; it only decides which part carries it, and correcting a deficit at one part without adding resource moves it to another. The exceptions are few and can be named in advance. We state this as a conservation law and derive it from a general model of how systems that must persist ration finite resources. The model's architecture, a regulator ranking access to a shared flow and parts going short in reverse rank, has been built separately in at least four fields: organism energetics, the Selfish Brain account of human metabolism, cell bioenergetics and plant carbon allocation. We found none carried beyond its own domain. We reached the same architecture from principles about institutions and state it once for all of them. From the general form follow results none of these models states formally, including why a steady reading carries no information about strain beneath it, the reserve left at the point of failure, when the speed of a decline causes lasting damage, and when rerouting overloads a route despite spare capacity. Of two pre-registered, blind-adjudicated tests, a predicted early warning before haemorrhagic decompensation failed. The prediction that the order of loss follows access documented in advance was supported, at reduced weight, in English council spending, though the version tied to the depth of funding cuts was not. We state what would refute the model.

1. Introduction

Systems that persist under finite resources show a recurring pattern. When resources run short, the output that matters most is held steady, while parts that matter less are run down out of sight. A bleeding patient's blood pressure holds while gut and kidney are starved of flow; a starving animal's brain keeps its mass while the body loses a third of its own; a council under austerity keeps its statutory services funded while discretionary ones are cut; a tree in drought stays green while its water-conducting tissue fails. The visible figure says little about the strain beneath it, until it breaks.

The law. Under scarcity, allocation is zero-sum. No ordering of access reduces the total shortfall; it only decides which part carries it. The exceptions are few and can be named in advance: adding

resource, lowering what parts require, and opening a gate that holds resource back. Drawing a store moves the shortfall in time, and drawing from outside moves it across the boundary.

As algebra, this is an accounting identity (Proposition 1). Kleinrock's (1965) conservation law for queues is one too: in a work-conserving single-server queue, no non-preemptive priority discipline changes the load-weighted sum of waiting times across classes; it only moves delay between them. The content of such a law is not its algebra. It lies in three places:

- **the reference:** each part's requirement is fixed in advance and never lowered because the part has shut down, so load cannot be defined away;
- **the exceptions:** they are named and can be checked;
- **practice:** how routinely it acts as if the law were false, by treating the visible figure.

Four received views the law contradicts under scarcity.

- **"Correcting an abnormal reading treats the problem."** In a resource-short system, correcting one figure without adding resource moves the deficit to another part (Proposition 1).
- **"A stable reading means a stable system."** The reading is held by spending stores and lower-ranked parts. While they last it carries no information about the load beneath it, and it breaks only when they run out (Propositions 2 and 3).
- **"Resilience is a property a system has."** Under scarcity, resilience is a store being spent and a lower-ranked part going without. It is borrowed from the system's own parts (Proposition 1).
- **"The failing organ is the problem."** Only sometimes, and the model says when. If lower-ranked parts are still supplied when an organ fails, the cause is its supply line or the organ itself (Proposition 7). If they have already gone short, the failure is the last visible step of a system-level allocation: look down the order (Proposition 4).

What is new. The same architecture has been built at least four times. Organism energetics (Kooijman 2010), the Selfish Brain account of human metabolism (Peters et al. 2004), the hierarchy of ATP consumers in cells (Buttgereit and Brand 1995) and sink priority in plants (Minchin, Thorpe and Farrar 1993) each describe a ranked order of draws on a shared flow, with the lowest-ranked going short first. Each was built for one domain, and our searches found none carried into another. Where the order was measured (in cells), emerged from mechanism (in phloem transport) or survived pre-registered review (the brain under caloric restriction), it is a property of the systems, not a modelling convenience. We arrived at the same architecture from a fifth direction, principles about conservation and priority in institutions, and state it once for all of them, as the Persistence Allocation Model. Three things none of them carries:

1. **an order of loss fixed in advance from documented properties of access** (pathway constriction, autoregulation, affinity, statutory status), never read off the observed order;
2. **a conserved ledger of load**, which counts unmet requirement against a fixed reference and records where every unit of it lands;
3. **application outside bodies.**

Of the results we derive from the general form, eight are stated formally by none of them (Table 3).

Tests. We report two pre-registered held-out tests, each replicated and adjudicated blind. One failed; one gave partial support, in the model's first application outside bodies (Section 5).

This paper:

- Section 2: the four instances, and theories that state parts of the architecture in words;
- Section 3: the model in formal terms, with eleven propositions;
- Section 4: the predictions, and how they separate the model from its neighbours;
- Section 5: the tests;
- Section 6: scope, limits and failures;
- Section 7: what follows.

2. Four instances of one architecture

Organism energetics. Dynamic Energy Budget (DEB) theory is the architecture for whole organisms (Kooijman 2010). It ranks the uses of reserve, paying somatic maintenance before growth and reproduction (the κ rule). It releases reserve in proportion to its content, shrinks structure when reserve cannot pay maintenance, and treats defence as "more facultative" than somatic maintenance. It conserves mass and energy. It ranks uses, not organs; it keeps no ledger of unmet requirement; and it allocates no repair, since damage in DEB is irreparable. We import four of its forms (Section 3.2).

The brain. The Selfish Brain theory is the architecture in human metabolism, with the brain as both the protected consumer and the regulator of the body's energy supply (Peters et al. 2004). The brain gives priority to its own supply by inhibiting glucose uptake into muscle and fat. In later formulations it suppresses insulin, closing the insulin-dependent route into muscle and fat while drawing through an insulin-independent one. This is set out as an energy-conserving supply-chain model (Peters and Langemann 2009), and a brain-centred compartment model has been analysed formally (Göbel and Langemann 2011). Its order has survived pre-registered systematic review: under caloric restriction the brain lost almost no mass while the body lost a great deal (Sprengell, Kubera and Peters 2021a). It is formal for two compartments and one resource. It has no order fixed in advance across many parts and no ledger.

The cell. In thymocytes the architecture was measured. A hierarchy of ATP consumers loses supply in order: macromolecule synthesis first, ion pumping later, proton leak last (Buttgereit and Brand 1995). It is a measurement, without a general model.

Plants. In phloem transport the architecture emerges from mechanism: "sink priority" follows from the transport network and the kinetics of the sinks (Minchin, Thorpe and Farrar 1993). Hierarchical plant growth models instead assume strict priority among organs (Grossman and DeJong 1994; classified as hierarchical by Marcelis and Heuvelink 2007). The order is assumed in two places, hierarchical plant models and DEB, where maintenance priority is a postulate. It was measured in cells, emerges from mechanism in phloem, and has been tested in the brain.

Parts of the architecture stated in words.

- The energetic model of allostatic load (Bobba-Alves, Juster and Picard 2022) proposes that the energetic cost of stress first uses up reserve capacity, then squeezes growth, maintenance and repair, potentially without raising total energy expenditure. That is the model's hidden load and its repair cut first, stated in words.
- Triage theory (Ames 2006) proposes that scarce micronutrients go to proteins needed for short-term survival over those needed for long-term health, partly through binding affinity, at the level of enzymes, cells and organs.
- A unifying theory of hypoxia tolerance (Hochachka et al. 1996) describes a balanced suppression of energy supply and demand: ion pumping and protein synthesis are cut back (channel and

translational arrest) while the cell's energy state is held.

- Allostasis (Sterling 2012) treats regulation as predictive and brain-led.
- The central governor model of exercise (Noakes 2012) holds that exercise is stopped before any system fails, with a reserve of motor units always kept. It is contested: Shephard (2009) finds convincing experimental evidence for its corollaries lacking, and argues that a plateau in oxygen consumption at exhaustion counts against a limiting central governor.
- The selfish immune system (Straub 2014) adds the immune and repair system as a second claimant that can take control of the body's spare energy. In this model, the brain and immune settings are two modes of one governor function.
- Perceptual control theory (Powers 1973) describes hierarchies in which higher levels set the reference values of lower ones. It gives the governor its form, but has no resource, store or ledger.

What the convergence shows. These fields developed in separate literatures; none of the four cites another's allocation model. Their agreement is evidence that the architecture is real. It is not evidence that the results derived from it here are true: those stand or fall by test (Sections 4 and 5).

Table 1 condenses the comparison; the full table, with eighteen theories, is Supplement S3.

Table 1. Prior theories against the model's components (F, formal; W, stated in words; E, measured, without a general model; p, partial or different form; blank, not found in what we read).

Theory	Governor	Order from access	Order fixed in advance	Stores drawn first	Repair cut first	Load ledger	Outside bodies
Dynamic Energy Budget	p	p		F		p	p
Selfish Brain	F	F (two compartments)	p	F		p	
Allostatic load (energetic)	W	W		W	W	W	
Triage theory		W	p	W	W	W	
Plant allocation models		F		p		p	
Cell ATP hierarchy		E (measured)			E		
This model	F	F	F	F	F	F	F

3. The model in formal terms

The general model, the Persistence Allocation Model, states the architecture as follows.

In a system that regulates its own persistence, a governor holds the levels its persistence depends on by regulating access to finite shared resources among parts that have no demand of their own; the allocation among parts is the resulting flow. Each part works to the limit of the scarcest resource that reaches it. When what reaches a part falls short of what the reference state requires, whether supply falls, requirement rises or access is restricted, load appears at that part. Where the system as a whole is short, the governor draws its stores, and lower-ranked parts lose access first and switch units off. The routine output, the record, holds until nothing more can be taken: the record sees compromise, not stress. Load is relocated, never removed: the resource gap is met from stores, met from outside the

boundary, or left unmet at a named part, and unmet load leaves its residue in switched-off, lost or scarred units, or in work not done, which may land across the boundary. A part scales down without harm when supply falls no faster than it can switch units off; units are lost when supply falls faster; the part is scarred only when those losses destroy what rebuilds it. Repair is its own network, governed like any part: under a sustained shortfall it loses access and lost units wait. Recovery runs the other way: the intake first, then parts and stores in order of value, with stores first when the system has learned its world is scarce. A part dies only when its route back is cut; the system collapses when load reaches the top or a non-bypassable link is cut, and dies only when no route back remains.

3.1 Architecture

Time runs in steps. In each step three functions apply:

$$\begin{aligned} g(t) &= G(\text{sensed state}(t), \text{environment}(t)), \\ a_{ir}(t) &= \Phi_{ir}(\{U_r, \text{stores}\}, \text{network}, g(t), s_i(t)), \\ s_i(t+1) &= F_i(s_i(t), \{a_{ir}(t)\}_r). \end{aligned}$$

- **The governor** G sets access, g : store release, intake, the capacity of pathways and gates, and the repair network's access. It does not set allocations, and it does not switch units. It is a function, not necessarily a place: a signal produced inside a part belongs to it.
- **The network** Φ turns access into the realised flow a_{ir} of resource r to part i : physical in bodies (pressure over resistance; transporters), a budget in organisations.
- **A part** F_i changes state only through what reaches it. It has no utility, claim or demand. Damage from outside enters as a loss of units or of pathway capacity.

A part consists of K_i units, each **active**, **switched off** (its route back kept, basal maintenance only, little or no work), **lost** (rebuildable) or **scarred** (lost with its template). One part, the **top**, produces the routine output whose level the governor protects; the ratio of that output to its level is the **record**.

3.2 The reduced form

Where the network is not modelled explicitly, the flow is written as an ordered draw. For resource r the available flow in a step is $S_r = U_r + I_r + \sum_s d_s$: supply through the intake, resource drawn in across the boundary, and store draws. Parts draw from it in phases, and within a phase in a fixed **order** π_r , each up to its draw:

1. the top's full need;
2. every other part's basal maintenance;
3. support parts' work and renewal (parts the top depends on, and the intake);
4. every other part's work and renewal, with work cut by economising;
5. what is left: reactivation and rebuilding against refilling the stores, by marginal value.

What a part does not draw stays in the flow; nothing is returned.

Rank. π_r is the order in which parts lose adequate access to r when r alone is scarce. **It is fixed in advance from documented properties of access** and never read off an observed order of loss.

Work, stores and units.

- **Work** is limited by the scarcest resource, $w_i = \min(\hat{c}_i, (1 - \epsilon_i)w_i^0, \min_r A_{ir}^w / \kappa_{ir}^w)$, with $\hat{c}_i$ effective capacity and w_i^0 reference work. The synthesising-unit form (Kooijman 2010) is the smooth general case.
- **Renewal** needs a baseline per active unit plus wear per unit of work, $\kappa^n n^a + \kappa^u w$.
- **Stores** release in proportion to content by default, $\rho_s = k_s L_s$ (DEB), or at a constant rate until empty. A fraction of undrawn flow refills them; the rest is spilled.
- **Units change only through supply.** Unpaid basal maintenance is paid from the part's own units, so the part shrinks (DEB). Unmet renewal switches units off at up to θK per step, and the rest fail at rate $1/\tau_f$. Losses in one episode beyond a template limit $Q^* K$ are scarred.
- **Repair** is the work of a set of parts, the repair network, governed like any other and serving damaged parts strictly by rank.

3.3 Load and the two ledgers

For each part and resource, the **reference allocation** q_{ir}^0 is fixed in advance as a rule and never lowered because units have switched off. **Load** is its unmet part, $\ell_{ir} = [q_{ir}^0 - a_{ir}]_+$. It has three origins: falling supply, rising requirement, or a governor setting (a mode) that restricts access while resources suffice. The last is admitted only for a setting and gate documented independently in advance.

The resource ledger. With the gap $\Gamma_r = \sum_i q_{ir}^0 - U_r$, every unit of the gap is carried by a store, met from outside, or left unmet at a named part (Proposition 1).

The state ledger. It records what unmet load leaves behind: units switched off, lost or scarred, and work not done, which may land on another system. Residue is a consequence of load and is never added to the resource ledger.

Assumptions for the results:

- results hold resource by resource;
- the ordered draw above, with a fixed order;
- the store release forms above;
- a reference fixed in advance;
- units changing only through supply, except by outside damage.

The margin M is what can go unmet below the top before the record moves. It is ordinary parts' draws D_4 if the top depends on its supports, and D_4 plus support parts' draws P and basal maintenance B if it does not. N is the top's full need.

3.4 Propositions

Proofs are in Appendix A. Six further results (a warning lead time, the limit under conflicting orders, shrinking, economising, the fuse, rising requirement) are in Supplement S1. **Every quantitative statement was checked numerically** against an independent implementation of the reduced form (Supplement S2). Two of the checks corrected earlier statements of the same results, and review corrected two more, each then confirmed numerically.

Conservation

Proposition 1 (relocation under scarcity, creation by gating). In any step, $\sum_i \ell_i = \Gamma - \sum_s d_s - I + R_{\text{unused}} + X$, where R_{unused} is resource left unused (refilled, spilled or left in the flow) and

X is allocation above reference.

1. **Scarcity-limited regime.** Where the flow is fully used and no part receives above its reference, total unmet load is the same under every access ordering at fixed supply, store draw and outside input. Access decides only where it lands.
2. **Access-limited regime.** Where a gate leaves resource unused, total unmet load rises one for one with the resource left unused, balanced by refill or spill. Opening the gate removes that load without any other part losing.

The identity is elementary. Its content is the reference fixed in advance and never lowered, and the named exceptions: resource added, requirement lowered, or a gate opened (Section 1). In a resource-short system, restoring one part's supply without adding resource leaves an equal deficit elsewhere. In an access-limited system (growth signalling holding a maintenance process shut while nutrients are present), opening the gate relieves the process with no deficit elsewhere. The two regimes predict different consequences of the same intervention.

The silence and the break

Proposition 2 (silence). The top's draw is met if and only if $S \geq N$; every part's basal maintenance and every support part's draw is met if and only if $S \geq N + B + P$. Hence the record is flat this step and the next if and only if $\Gamma \leq \sum_s d_s + I + M$.

Inside that region the record carries no information about load below the top. Outside input extends the silence exactly as a store does. It breaks at two thresholds: at once, and one step later through dependency.

Proposition 3 (store left at the break). A constant gap Γ per step is carried by one store from level L_0 , with no outside input, and $M < \Gamma \leq kL_0$. Under full release the record breaks when the store is empty, whatever Γ . Under proportional release it breaks when the store falls to $L^* = (\Gamma - M)/k$: **a faster shortfall leaves more of the store unused**, linearly, with slope $1/k$.

The intuitive rate-independence of the break holds only for full release. Under the default release, the reserve remaining at decompensation should rise with the rate of loss.

The order of loss

Proposition 4 (order). Within a phase, the parts short of their draw form a lower segment of π . No part loses basal maintenance while any part draws for work. The top goes short last.

Strict priority of this kind is an assumption in hierarchical plant models (Grossman and DeJong 1994; Marcelis and Heuvelink 2007). Here it is the reduced form of an access network, with the order fixed in advance.

Proposition 5 (rank under saturable uptake). Parts take a shared resource by saturable uptake, $v_i = V_i C / (K_i + C)$, from a pool at level C . Let $r_i = q_i^0 / V_i$ be each part's reference requirement as a share of its maximum uptake. Part i is adequately supplied if and only if $C \geq C_i^* = K_i r_i / (1 - r_i)$, so parts lose adequate access in descending order of C_i^* . Affinity (K_i) alone decides the order only where r_i is equal across parts. Priority is strict where neighbouring thresholds are far apart, and shared where they are close.

Rank is then a measurable property, fixed before outcomes. It combines affinity, capacity and requirement: a high-affinity part working near its maximum can lose adequate supply before a low-affinity part with large capacity and a small requirement. This connects the model's rank to

Michaelis-Menten competition, to measured hierarchies of ATP consumers (Buttgereit and Brand 1995) and to triage by binding affinity (Ames 2006).

Harm

Proposition 6 (rate decides harm). For a renewable part with basal maintenance met, let the units the repair network can renew fall by φ per step to a depth Δ .

1. If $\varphi \leq \theta K$, no units are lost at any depth.
2. If $\varphi > \theta K$, the loss satisfies $\Delta(1 - \theta K/\varphi) - \tau_f(\varphi - \theta K) \leq \Lambda \leq \Delta(1 - \theta K/\varphi)$.
3. A scar is possible only if $\Delta(1 - \theta K/\varphi) > Q^*K$, and certain if the lower bound exceeds Q^*K .

Depth without speed never harms; speed without depth never scars; a scar needs both, and the threshold depth falls as speed rises. Fixed capital ($Q^* = 0$) is scarred by any fall faster than its switch-off rate.

Failure

Proposition 7 (exhaustion, severance and constriction). Let delivery to the top and its supports be bounded by pathway capacity, the minimum cut between source and part (Ford and Fulkerson 1956). If the record falls while some ordinary part still draws for work, or a store still has release headroom, then either the minimum cut to the top or a support part is below its need (severance at zero, constriction above zero), or damage from outside has removed units of the top or a support part.

Failure while willing receivers are still supplied identifies a cut or narrowed link, or outside damage. Each is observable independently.

Proposition 8 (rerouting). A pathway with parallel routes loses route e . If its flow is reallocated where there is headroom, no surviving route is overloaded if and only if the surviving headroom covers the lost flow. If flow divides by a physical rule (in proportion to conductance), a surviving route can be overloaded even when total headroom suffices. Where a collateral joins two parts' branches, flow reaching one through it is taken from the other at fixed supply (steal).

Cascades and steals are expected in physically divided networks (vessels, pipes, power lines) even with spare capacity, and in networks reallocated by choice (budgets, routers) only when total capacity is short. Subclavian steal and line-outage redistribution are instances of the first; a budget moved to a channel with room is the second.

Proposition 9 (collapse, not death). A system outside its viable set is inside the capture basin (Aubin 1991), so in collapse rather than death, if:

- supply, including outside input, can return to and be held above the top's need, every surviving part's basal maintenance and, where the top depends on them, the support parts' draws, with a positive remainder for the repair network;
- the top is not scarred below what the record requires;
- every non-bypassable link the record depends on has capacity above zero or can be restored;
- every part the record depends on has its template intact and a pathway for repair;
- the repair network has positive capacity, or can itself be restored.

The time to re-enter the viable set is at least the longest rebuild time among those parts. The conditions are measurable before the outcome.

Recovery and repair

Proposition 10 (recovery order and partial refill). The intake comes back first after refeeding, whatever its rank. Thereafter surplus goes by marginal value. With F the distribution of remembered episode depths, c_S the cost per unit of a store being short, ρ the expected frequency of shortfall and V the best competing value of a part's binding units, the store refills to $L^* = F^{-1}(1 - V/(c_S\rho))$ if $V < c_S\rho$, and not at all otherwise.

Stores refill only part way while parts still bind, deeper after frequent or deep shortfalls. After short episodes parts come back first; after long ones, stores do. The refill level is the newsvendor critical fractile; the allocation rule is marginal analysis of spares (Sherbrooke 1968).

Proposition 11 (repair under strict rank). With repair work W per step binding and damage D_j to parts in rank order, part i finishes healing at step $\lceil \sum_{j \leq i} D_j / W \rceil$. The highest-ranked damaged part heals as fast as alone; each lower-ranked part is delayed by the damage ranked above it. Under a sustained shortfall the repair parts lose access, so every finishing time rises.

Strict-rank repair is distinguishable from shared repair: under strict rank, the most important damaged part does not slow when others are damaged too.

4. Predictions, and how they separate the model from its neighbours

The model's predictions fall into two layers. **Layer 1** holds for any feedback loop with a finite stock: the record is flat while a buffer is drawn (G1), the break comes at a set depletion, and a larger buffer gives a longer silence. Confirming layer 1 says little about the model. **Layer 2** needs selection or design: the order of loss, switches, the order of recovery, repair competition. Table 2 lists the critical and supporting predictions; the full list, with status, is Supplement S4.

Table 2. Main predictions.

No.	Prediction	Derived from	Status
G1	The record stays near normal while lower parts and stores move	Prop. 2	Natural observation: consistent; H1 check not consistent (low weight)
G2	Lower-ranked parts lose access first, in ascending rank	Prop. 4	Natural observation: consistent
G3	With a store carrying the gap and its release not binding, the break comes at the same cumulative shortfall whatever the rate. Under proportional release, a faster shortfall leaves more of the store unused at the break	Prop. 3	Not yet tested; sheep and trees fit the first clause (Table 4)
G8	Rate decides harm; a scar needs speed and depth	Prop. 6	Compatible; not yet tested
G12	Recovery from small knocks slows before the break where release headroom shrinks	Section 3; Supplement S1	Failed its first held-out test (Section 5.1)
G18	Parts sharing a loaded dependency move together before the break; parts sharing only rank do not	Section 3.3	Not yet tested
G20	Failure with willing receivers supplied means a cut or constricted link, or outside damage	Prop. 7	Not yet tested
G23(b)	Under strict-rank repair, the highest-ranked damaged part heals as fast as alone	Prop. 11	Not yet tested

No.	Prediction	Derived from	Status
G24	Cascades and steals along substitutes where flow divides by physics, even with spare capacity	Prop. 8	Known in power systems; not yet tested as a cross-domain prediction
G25	The order of loss is predicted by properties of access documented beforehand	Props. 4, 5	Supported in its first held-out test; the scarcity version not supported (Section 5.1)
G26	Stores refill only to a critical fractile while parts still bind	Prop. 10	Not yet tested

What the model predicts that neighbouring theories do not state. Table 3 lists results that follow in the model but are not stated in the neighbouring theories as we have read them. Some appear in words (in allostasis especially), and rerouting by physics is known in power engineering; none is stated formally across domains. Silence is not contradiction: the entries mark where the model makes a claim those theories do not make, and where a test can tell them apart.

Table 3. Results not stated in neighbouring theories.

Result	DEB	Selfish Brain	Allostasis and allostatic load	Control theory
Scarcity load is relocated; gating creates and removes load (Prop. 1)	Conserves mass and energy; no ledger of unmet requirement	Build-ups in front of bottlenecks; no ledger across many parts	Hidden load, in words	No resource ledger
The record carries no information inside the silence region (Prop. 2)	Not stated	Brain held while body loses, two compartments	Stated in words	Not stated as a shortfall budget
Store left at the break rises with the rate of shortfall (Prop. 3)	Gives the release form; not the break	Not stated	Not stated	Not stated
Rank under saturable uptake from affinity, capacity and requirement (Prop. 5)	Not stated across parts	Access by insulin dependence, two compartments	Not stated	Not stated
Scar threshold in speed and depth (Prop. 6)	No unit loss or scar; damage irreparable	Not stated	Wear, in words	Not stated
Failure with receivers supplied implies a cut or narrowed link, or damage (Prop. 7)	Not stated	Not stated	Not stated	Not stated
Physical against chosen rerouting (Prop. 8)	Not stated	Not stated	Not stated	Known in power-systems engineering; not stated across domains
Repair served strictly by rank (Prop. 11)	No repair allocation	Not stated	Repair squeezed, in words	Not stated

The sharpest discriminating tests are these:

- **Relocation:** in a resource-short system, a matching deficit should follow any correction that adds no resource.
- **The order of loss from access** (G25) in systems the model was not built on.
- **The scar threshold:** harm needs speed and depth together.
- **Strict-rank repair** (G23 b).

5. Tests so far

5.1 Held-out tests

Both tests were pre-registered before any outcome data were opened. Each computation was replicated by a second AI system (ChatGPT, OpenAI): it wrote its own analysis script from the pre-registration and our code, without seeing the data or the results, and the script was run on the same files. Each verdict was then given blind against fixed adjudication rules by the same system, in a separate session with no shared context. Records are in Supplement S5.

H1: warning before decompensation in surgical blood loss (G12).

- **Data:** VitalDB, an open database of high-resolution intraoperative recordings (Lee et al. 2022).
- **Prediction:** before falls in arterial pressure in cases with high blood loss, recovery from small fluctuations should slow. This was measured as a rise in the lag-1 autocorrelation of mean arterial pressure, against controls from the same case.
- **Cohorts:** the primary cohort (no fluid boluses) had no qualifying high-loss cases. The fallback cohort, with a loss threshold of 15% of estimated blood volume, gave 25 high-loss and 42 low-loss cases.
- **Result:** the median excess rise was -0.072 (one-sided $p = 0.48$), and high-loss cases did not exceed low-loss ones ($p = 0.78$). **G12 failed**, at half weight, with two qualifications:
 - the test is runnable only under the pre-data reading of missing 10-second bins;
 - the step-down of the loss threshold inside the fallback cohort was fixed in code before the data were opened, but not stated in the pre-registration.
- **G1 check (low weight):** pressure stayed within 20% of baseline while haemoglobin fell in 10.7% of 75 cases, against a pre-stated "more than half". Not consistent.
- **Reading under the model's standing rule:** the first candidate for revision is the mapping (the store's release profile under anaesthesia, where an anaesthetist acts as an outside loop holding pressure), not a new mechanism. One failure does not remove G12. Any explanation must name a specific fault and be tested in advance.

PT1: order of loss from documented access in English local government (G25).

- **Data:** single-tier councils in England, 2014-15 to 2019-20: 121 councils and 93 spending lines.
- **Access property:** each line was classified **blind**, before any spending data were opened, by whether a statutory duty attaches to it: class A, a duty; class B, a duty of uncertain level; class C, discretionary.
- **Primary question (fixed against dynamic priority): inconclusive** at full weight. Projected client growth published before each budget showed no detectable effect on which lines were protected ($\beta = 0.05$; 95% interval -0.75 to 0.86), but the interval includes the smallest effect set in advance as meaningful (0.25).
- **G25: supported, at half weight** (the broad pattern was known in advance). Within a council and year, real spending per head on class A lines grew about 4.6 percentage points a year faster than on class C lines (one-sided $p = 2 \times 10^{-11}$). Class B was not distinguishable from class C, and that step fails its ordering when London is excluded.
- **G25-C: not supported, at half weight.** The scarcity version, that the gap widens where funding fell more, failed: the interaction was 0.22 (SE 0.25).

This is the model's first application outside bodies, and its first held-out support, for order of loss only.

5.2 Natural-system observations

Before the held-out tests, the model was checked against published findings in several natural systems, with predictions written down before the sources were opened. These checks are **not tests**: some outcomes were known in advance, sources were often read through summaries, and three early refinements of the model were built from blood-loss and fasting findings (Section 6). We report them as compatible observations (Table 4; full records in Supplement S6).

Table 4. Natural-system observations (compatible, not diagnostic).

Feature	Blood loss	Plants in drought
Record held while load rises	Arterial pressure well maintained while cardiac output falls (Evans et al. 2001); vital signs stable in early bleeding while the compensatory reserve falls (Convertino et al. 2016)	Changes in foliage colour lag hydraulic failure, best predicting trees already dead rather than dying (loblolly pine saplings; Hammond et al. 2019)
Lower priority pays first	Splanchnic vasoconstriction the dominant compensation in a model calibrated to 35 adults under lower-body negative pressure (Bergauer et al. 2026); renal resistance rising before carotid in a model calibrated to 43 swine (Sadid et al. 2026, preprint)	Not used here: the leaves-first ordering is contested (Section 6)
Buffer behaves as a stock	Blood volume removed at a 30 mmHg fall in mean pressure: $27.0 \pm 4.2\%$ at about 0.4% of blood volume a minute against $27.3 \pm 3.2\%$ at about 2% a minute (mean $\pm$ SE; 8 sheep, crossover; Scully et al. 2016)	Saplings died at the same loss of conductivity under fast and slow drought, with species-specific thresholds (about 95% and 45%; Dai, Wang and Wan 2018)
Break at a set depletion	Decompensation (sympathetic withdrawal) once cardiac output falls to 50 to 60% of rest, about 30% blood loss (Evans et al. 2001)	About 80% loss of conductivity in loblolly pine saplings (Hammond et al. 2019)

The sheep result fits full release, or a store with turnover high relative to the gap (Proposition 3). It is an observation about the release profile in that setting, not a test.

6. Scope, limits and failures

Scope.

- **Persistence is a viability constraint, not something maximised.** The governor keeps the system within the states compatible with its continued existence; within that envelope the system pursues its purposes.
- **Terminal reproductive programmes are out of scope** (semelparous salmon, some marsupial males, monocarpic plants). There, individual persistence stops being the protected constraint. The model should work before the reproductive switch and fail after it.
- **The model is not a theory of action selection.**
- **Boundaries, protected levels and the classification of components are fixed before outcomes** and are not redrawn after a counterexample.

Limits of the formal results.

- The propositions hold for the reduced form, not for a general network. Where several complementary resources bind together and their orders conflict, ordinal ranks do not decide the outcome (Supplement S1): the network must be mapped explicitly.

- The order of loss is assumed fixed. **Whether it is fixed or reverses with conditions is open:** PT1's primary question was inconclusive (Section 5.1). A valid test needs two recipients competing for the same scarce resource at the same time, with the governor's mode held constant.
- The functional forms not taken from DEB (switch-off and failure rates, the scar rule) are modelling choices. No parameter has been estimated from data.

Failures and findings against the model.

- **G12 failed** its first held-out test, and **G25-C was not supported** (Section 5.1).
- **Findings logged against the model, not yet weighed:**
 - autopsies after prolonged inanition (Krieger 1921, as cited by Peters and Langemann 2009) found heart, liver, pancreas and kidney all losing about 40% of their mass while the brain lost under 2%, with no order among the organs below the brain. Whether this conflicts with the model depends on the order fixed for energy and protein at a starvation mapping;
 - in 12 Australian tree species, the predicted leaves-before-stems order of hydraulic failure (vulnerability segmentation) was "universally absent or negative" (Peters and Choat 2025).
- **Logged, explained in advance and checked:** in mature spruce, an earlier drought eased physiological stress in a later one (Hikino et al. 2026), where a stressed start should fare worse. The explanation named before checking, that the first drought cut leaf area and so the demand on the same water supply, holds: leaf area fell by 60% and was still 30% lower four years later. The finding is not counted against the model.

Details and weights are in Supplement S7.

Independence. The convergence in Section 2 rests on four fields that developed separately and does not depend on our route. Our own route is not independent of them:

- the AI system that collaborated in building the model was very likely trained on these literatures;
- some components were already known to us when they were formed;
- three early refinements were built from blood-loss and fasting findings: shared upstream supply, a threshold switch, and the point at which a store begins to release.

The dated record is in Supplement S11.

7. Discussion

Mapping a system. The model's main methodological claim is that the order of loss can be fixed before outcomes. Mapping a system (Supplement S9 in full) means fixing, before any outcome data are opened:

- the boundary, the protected level and the viable set, and whether outside support counts;
- the resources, whether they are complementary or substitutable, their stores and release profiles;
- the governor's levels and what it acts through;
- each part's units, rank for each resource from documented properties of access, requirements, reference allocation, rebuild time and template limit;
- the repair network;
- the pathways, and whether flow among routes divides by physics or by choice;

- any modes and gates invoked for access-limited load, each documented independently;
- the clock.

Anything chosen with the expected outcome in mind is declared fitted. **Where access is by saturable uptake, the rank is the order of adequacy thresholds** (Proposition 5), computable from measured affinity, capacity and requirement.

What would count against the model:

- in a system resource-short by its own mapping, an intervention that restores one part's supply without adding resource and without a matching deficit appearing elsewhere;
- in a held-out system, an order of loss that contradicts the order fixed in advance from documented access (G25);
- harm without speed: units lost or scarred under a fall in renewal slower than the documented switch-off rate (Proposition 6);
- an ordinary system, in ordinary conditions, whose own regulation destroys recoverable viability with no higher-level system being protected.

A fifth received view. Homeostasis is usually described as balance. Under scarcity, holding the top steady is paid for further down: the steadiness of one level is the shortfall of another (Propositions 1 and 4). Allostasis says the same of predictive regulation in words (Sterling 2012); the ledger gives it a count.

Next tests. Every prediction not yet tested can be tested; none is untestable in principle.

Prediction	What a test needs	Where the data are
G25, order of loss from access	A held-out system not known in advance to the author or the AI collaborator, with access documented before outcomes	Public budgets outside PT1's frame; hospital bed allocation; plant carbon labelling
G12, warning before the break	A test of the store's release profile in blood loss, with the mapping fault behind the first failure named in advance	A haemorrhage source not opened in H1 (graded-rate animal haemorrhage series; MIMIC waveform records of bleeding patients). Any retest in VitalDB counts at reduced weight
Fixed against dynamic priority	Two recipients competing for the same scarce resource at the same time, with the governor's mode held constant	Council spending outside PT1's frame (English single-tier councils, 2014-15 to 2019-20): a later English period, or Scottish or Welsh councils; organ perfusion studies
G3, the break and the rate of shortfall	The break defined in advance, at two or more rates, with the store's release measured	Graded-rate haemorrhage; fast and slow drought in one species
G8, rate decides harm	Paired sudden and gradual losses of the same route	Stroke and coronary occlusion series; plant girdling
G18, co-movement before the break	Multi-organ time series, with dependencies mapped before opening	ICU databases (MIMIC-IV, eICU; credentialed)
G20, failure with receivers supplied	Network outages with routes known	Grid outage records; arterial stenosis series
G23(b), strict-rank repair	Repair measured by part after multiple injury	Trauma registries
G24, physical against chosen rerouting	A matched pair of physically divided and budgeted networks	Grid flow redistribution; budget reallocation
G26, partial refill	Store and shortfall records over repeated episodes	Local authority reserves; seasonal fat stores in wildlife

A proposed extension treats components with their own access loops (a tumour; a person in an institution) as nested systems; its first test, whether a tumour that only drains a host matches a host whose supply is cut by the same amount, is future work.

How this work was done

The model was built in October 2026 by one author working outside academia, with an AI system (Claude, Anthropic) as collaborator for the mathematics, literature searches, numerical checks and drafting.

- **Time-stamped record:** every version of the model, every prediction and every test rule is recorded, with its commit history, in a public repository archived on 7 October 2026 (Miller 2026; <https://doi.org/10.5281/zenodo.23222615>). Its full commit history is preserved in Software Heritage (swh:1:snp:cbdaf9b4e6c27ea5f2ce307d86ebcd49ca11fcf).
- **Predictions first:** each check's and each test's predictions were committed before its sources or data were opened.
- **Tests:** each held-out test was replicated by a different AI system (ChatGPT, OpenAI), which wrote its own analysis script without seeing the data or the results, and then adjudicated blind against rules fixed in advance by the same system, in a separate session with no shared context.

The dated record of the model's development is in Supplement S11.

Appendix A. Proofs of the main-text propositions

Proposition 1. Conservation within the step gives $U + I + \sum_s d_s = \sum_i a_i + R_{\text{unused}}$. With $\ell_i = [q_i^0 - a_i]_+$, $\sum_i (q_i^0 - a_i) = \sum_i \ell_i - X$. Hence $\sum_i \ell_i = \sum_i q_i^0 - U - I - \sum_s d_s + R_{\text{unused}} + X$, the identity stated. Regime 1: $R_{\text{unused}} = 0$ and $X = 0$, so the total contains no access term. Regime 2: $X = 0$ and R_{unused} equals the resource the gate leaves unused. ■

Proposition 2. The top draws first, then all basal maintenance, then support parts. So the top is met if and only if $S \geq N$, and basal maintenance and supports if and only if $S \geq N + B + P$. The top's work next step falls only through its own units (impossible while it is met) or through dependency on supports (excluded when they are met). Since $\sum_i q_i^0 = N + B + P + D_4$ and $S = U + I + \sum_s d_s$, $S \geq N + B + P$ is equivalent to $\Gamma \leq \sum_s d_s + I + D_4$; without dependency, $M = D_4 + P + B$. ■

Proposition 3. While $kL \geq \Gamma$ the store releases the whole gap, so L falls by Γ per step. Once $kL < \Gamma$ the store releases kL , L falls by the factor $(1 - k)$ per step, and $\Gamma - kL$ falls below the top. By Proposition 2 the record breaks at the first step with $\Gamma - kL > M$, that is $L < (\Gamma - M)/k$; in discrete time the level at the break lies in $((1 - k)L^*, L^*]$. Under full release with rate at least Γ , the store carries the whole gap until empty, independent of Γ . ■

Proposition 4. Draws are sequential within a phase: a part is short only if the flow is exhausted at its turn, so every later part draws nothing. Phases complete in order. ■

Proposition 5. $V_i C / (K_i + C) \geq q_i^0$ rearranges to $C \geq K_i q_i^0 / (V_i - q_i^0) = C_i^*$. The pool level falls monotonically with supply, so parts cross their thresholds in descending order of C_i^* . With r_i equal, $C_i^* \propto K_i$. With thresholds far apart, at most one part is near its threshold at any C (strict); otherwise several are short together (shared). ■

Proposition 6. Over the episode, switched-off and lost units together number Δ . During the decline, which lasts at least Δ/φ steps, the excess x of active over renewable units satisfies $x_{t+1} = (x_t - \theta K)(1 - 1/\tau_f) + \varphi \geq \varphi > \theta K$, so θK units are switched off each step and at least $\theta K \Delta/\varphi$ in all: hence the upper bound. At the end of the decline the excess is at most $\theta K + \tau_f(\varphi - \theta K)$, and no more can be switched off in the tail: hence the lower bound. If $\varphi \leq \theta K$, every unit not renewed is switched off within the step. ■

Proposition 7. If an ordinary part still draws for work, or a store has headroom, then $S \geq N + B + P$ (Proposition 2), so the flow could meet the top and its supports. If they are short, another bound binds: the pathway, whose capacity is the minimum cut, or their own capacity, which falls only through supply (excluded) or damage. ■

Proposition 8. Placing the lost flow into surviving headroom succeeds exactly when the headroom suffices; in a general network this is max-flow min-cut. By example: routes with capacities 10, 4 and 10 carry 5, 3.9 and 5. Losing the first leaves headroom 5.1 for flow 5, but a split in proportion to capacity sends about 1.43 to the second route, which then carries 5.33 against capacity 4. At fixed supply, conservation at a junction means flow leaving one branch through a collateral is subtracted from it. ■

Proposition 9. Under the first condition an access setting exists that fully funds the top, its supports and all basal maintenance (Proposition 2), so the top and its supports lose no further units (Propositions 4 and 6). Other parts may still lose units, but the record does not depend on them. The positive remainder funds the repair network, which under the remaining conditions rebuilds each needed part at its rebuild rate. The system therefore reaches the viable set in finite time and can stay there: membership of the capture basin. ■

Proposition 10. The intake is a support part, served in phase 3. A store's marginal value $c_S \rho(1 - F(L))$ is non-increasing in L ; greedy allocation by marginal value stops refilling where it falls to V , the stated quantile. Greedy allocation is optimal for separable concave value (Ibaraki and Katoh 1988). ■

Proposition 11. Under strict priority, cumulative repair work after t steps is tW , applied to damage in rank order; part i is complete when $tW \geq \sum_{j \leq i} D_j$. ■

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Supplementary material

S1. Supplementary propositions and proofs

Notation and assumptions as in the main text, Section 3.3. Main-text propositions are cited by their paper numbers.

S1.1 Warning before the break, and its lead time. Under proportional release ($\rho = kL$), with no outside input, and a constant gap Γ with $M < \Gamma \leq kL_0$:

- release headroom $kL - \Gamma$ falls linearly in time while the store carries the gap, and reaches zero at $L = \Gamma/k$. From then on, recovery from small perturbations of the record cannot draw on the store and slows;
- the break follows after a lead time $T_{\text{lead}} \approx \ln(\Gamma/(\Gamma - M))/(-\ln(1 - k)) \approx k^{-1} \ln(\Gamma/(\Gamma - M))$, which shortens as the gap grows relative to the margin and vanishes as the margin goes to zero;
- under full release, headroom is constant until the store holds less than one step's release, so there is no slowing before the store empties.

Proof. Headroom is zero at $L_w = \Gamma/k$. Thereafter L decays by the factor $(1 - k)$ per step until $L < (\Gamma - M)/k$ (Proposition 3); the number of steps is $\ln(L_w/L^*)/(-\ln(1 - k))$. ■

Note. This bears on G12, which failed its first held-out test (main text, Section 5.1). It is reported here, not in the main text, so that it is not read as a rescue: using it would require the margin and the store's turnover to be fixed in advance in a new test.

S1.2 The limit of rank under joint scarcity. Two parts each need one unit of each of two complementary resources per unit of work; one unit of each is available; the orders conflict (part A first for resource 1, part B first for resource 2).

- Allocating each resource by its own order gives neither part any work.
- Every split $(z, 1 - z)$ of both resources gives total work 1 and is Pareto efficient. The two ordinal orders do not choose among them.

Proof. Under the law of the minimum, allocation by order gives A both units of resource 1 and none of resource 2, and B the reverse. Any common split gives work z and $1 - z$. ■ Consistent with multi-resource allocation under Leontief preferences (Ghods A, Zaharia M, Hindman B, Konwinski A, Shenker S, Stoica I (2011), Dominant resource fairness: fair allocation of multiple resource types, Proceedings of the 8th USENIX Symposium on Networked Systems Design and Implementation). Where this case arises, the order of loss is set by the network, which must be mapped.

S1.3 Shrinking when upkeep goes unpaid. A part whose basal maintenance is short pays from its own units and loses $u = [\kappa^b n - a^b]_+ / (\kappa^b + m/y)$ of them, where m is the resource recoverable per unit broken down and $y \geq 1$ the overhead. The remaining units are exactly funded, and $u \leq n(1 - \beta)$ (the unfunded share), with equality when units hold nothing usable.

Proof. u solves $\kappa^b(n - u) = a^b + (m/y)u$. ■ This is the DEB shrinking rule with absolute preference for reserve (Kooijman 2010, Sections 4.1.5 and 3.7.4, eq. 4.6).

S1.4 Economising loses no units. Economising cuts ordinary parts' work access by a share ϵ ; it does not cut renewal. It therefore causes no unit loss at any speed. Its saving arrives at once for the

work and its wear, $\sum_i (\kappa_i^w + \kappa_i^u) w_i^0 \epsilon$ per step, and as idle units are switched off for their baseline renewal.

Proof. Units are lost only through unmet basal maintenance or renewal. Economising lowers the work draw, and with it the wear part of renewal need; the baseline need of units still active stays funded until they are switched off. ■

S1.5 The fuse. The lowest-ranked part is the first to go short (Proposition 4). It is the last to come back only if its marginal value in recovery is also lowest, since recovery allocates by marginal value (Proposition 10). The intake is the exception: it comes back first whatever its rank.

S1.6 Rising requirement. For a part with switched-off units, rising requirement is met first within active capacity, then by reactivation at up to $\theta_{re}K$ per step at a cost. Output falls short only when the rise outpaces reactivation, reactivation cannot be paid for, or requirement exceeds total capacity.

Proof. Work is bounded by active capacity; reactivation is the only route from switched off to active, and it is rate-limited and paid from what is left in the flow. ■

S2. Numerical checks

What was checked.

- Every quantitative proposition in the main text and in S1 was checked against an independent implementation of the reduced form (Section 3.2 of the main text).
- The implementation was written from the equations, not from the simulation engine used during the model's development.
- It covers one resource:
 - one step of the ordered draw (the top's need, then basal maintenance, then support parts, then ordinary parts, each phase in a fixed order);
 - the two store-release forms;
 - unit switch-off and failure;
 - saturable uptake;
 - strict-rank repair.

Design. Each check draws random instances from stated ranges and tests the proposition's statement on every instance. A single failure prints the counter-example. The seed is fixed (20261007), so every run is identical. The code uses only the Python standard library.

Result: every check passes.

Result	What is tested	Instances	Outcome
Proposition 1, scarcity regime	The ledger closes (gap = store draw + total load). Total load is unchanged when the order of basal and ordinary draws is randomly permuted, at fixed supply and store draw	2,000 (one top, six basal, two support and six ordinary draws; supply uniform up to total need)	Pass
Proposition 1, general identity	$\sum \ell = \Gamma - \sum d - I + R_{\text{unused}} + X$ with gates closed at random (probability 0.3) and allocation up to 1.5 times reference	2,000 (five parts)	Pass
Proposition 2	The top is met if and only if $S \geq N$; basal maintenance and support are met if and only if $S \geq N + B + P$	2,000	Pass

Result	What is tested	Instances	Outcome
Proposition 2, with outside input	In terms of the gap: basal maintenance and support are met if and only if $\Gamma \leq \sum d + I + D_4$, and the top if and only if $\Gamma \leq \sum d + I + D_4 + P + B$, with outside input I drawn at random	2,000	Pass
Proposition 3	Under proportional release the store left at the break lies in $((1 - k)L^*, L^*]$, with $L^* = (\Gamma - M)/k$. Under full release, less than one step's gap is left. A faster gap leaves more store unused (gaps 1.5, 2, 3 and 4). A gap at or below the margin never breaks	500 (L_0 50 to 200; k 0.01 to 0.2; M 0 to 5; gap between $M + 0.1$ and kL_0), plus fixed cases	Pass
Proposition 4	Within the ordinary phase, the parts short of their draw form a lower segment of the order. No basal maintenance is short while any work is drawn. When the top is short, nothing else is drawn	2,000	Pass
Proposition 5, strict limit	With four parts of equal capacity and half-saturation constants spaced by factors $R = 10^2, 10^4, 10^6$, saturable uptake departs from strict priority by at most 0.15, 0.02 and 0.002 at five supply levels. Shares fall with the half-saturation constant	Fixed cases	Pass
Proposition 5, adequacy order	Parts lose adequate supply in descending order of $C^* = Kr/(1 - r)$. Counter-example to affinity alone: $V = (1, 10)$, $K = (1, 10)$, $q^0 = (0.9, 0.5)$; the higher-affinity part loses adequate supply first	200 random three-part cases (near-ties, with $ \ln(C_i^*/C_j^*) < 0.05$, skipped), plus the counter-example	Pass
Proposition 6	A fall in renewable capacity no faster than θK loses no units. A faster fall loses between $\Delta(1 - \theta K/\varphi) - \tau_f(\varphi - \theta K)$ and $\Delta(1 - \theta K/\varphi)$	400 ($K = 100$; θ 0.02 to 0.2; τ_f 2 to 10; φ 0.2 to 40; Δ 5 to 90)	Pass
Proposition 8	When a route is lost, reallocating its flow into headroom overloads a surviving route if and only if total headroom is less than the lost flow. A split in proportion to capacity overloads a route in at least one case with enough headroom	2,000 networks (three to six routes, capacities 1 to 10)	Pass
Proposition 10	Refilling a store unit by unit while its marginal value exceeds the best competing value V stops within 0.2 of the critical fractile $F^{-1}(1 - V/(c_S\rho))$. There is no refill when $V \geq c_S\rho$	500 remembered episode depths (exponential, mean 20); $c_S = 10$, $\rho = 0.4$; five values of V	Pass
Proposition 11	Under strict rank each damaged part finishes at step $\lceil \sum_{j \leq i} D_j/W \rceil$. The highest-ranked damaged part heals as fast as alone	1,000 (five parts; W 0.5 to 5; damage 0 to 10)	Pass
S1.1	The lead time from the start of warning to the break is within one step of $\ln(\Gamma/(\Gamma - M))/(-\ln(1 - k))$	500 (ranges as for Proposition 3)	Pass
S1.2	Allocating each of two complementary resources by its own conflicting order gives no work. Every common split gives total work 1	Exact case	Pass
S1.3	After shrinking, the remaining units are exactly funded, and the loss is no larger than the unfunded share	2,000	Pass
S1.4	Economising at any speed leaves renewal fully funded. Its first-step saving equals $(\kappa^w + \kappa^u)w^0\epsilon$. Consolidation brings active units down to the economised work	300 (θ 0.02 to 0.2; economising depth 0.1 to 0.9; speed one to ten steps)	Pass

Not checked numerically. These follow directly from the definitions, so a numerical check would only restate them:

- Proposition 7, a statement about which bound binds, with the minimum cut given by max-flow min-cut;
- Proposition 9, a constructive condition for re-entering the viable set;
- S1.5 and S1.6.

The code also checks one result that is not used in this paper: a drain treated as a cut in supply. It belongs to the proposed nested-systems extension.

Corrections the checks and review made. Each was confirmed numerically after it was found:

1. **Proposition 6:** the loss expression $\Delta(1 - \theta K/\varphi)$ was first stated as an estimate. The checks showed it is an upper bound, and the lower bound was added.
2. **Economising:** an earlier statement implied that fast economising causes unit loss. The checks (S1.4) showed it does not: economising cuts work, not renewal.
3. **Proposition 1:** total load was first stated as invariant to access in general. Review showed this holds only where the flow is fully used and no part receives above its reference. The proposition now states two regimes, and the general identity (with $+X$, not $-X$, as first written) is checked.
4. **Proposition 5:** rank under saturable uptake was first attributed to affinity alone. Review showed it depends on capacity and requirement as well, and the counter-example above is now part of the check.
5. **Proposition 2 (outside review):** the silence condition left out outside input. It is now $\Gamma < \sum_s d_s + I + M$. With random outside input, the earlier condition gave the wrong answer in 622 of 2,000 cases; the corrected one passes all 2,000 (table above).
6. **Proposition 9 (outside review):** the first condition funded the top and basal maintenance but not the support parts the top depends on, nor the repair network, so it did not guarantee recovery. The condition now requires both, with a positive remainder for repair. A condition that the repair network has capacity, or can be restored, was added, and the proof was revised.

Faults in the checks themselves, fixed before the results above:

- the generator for Proposition 3 produced gaps below the margin, which cannot break, so those cases are now excluded by construction;
- the uptake solver's search range was too narrow for widely spaced half-saturation constants, so it now bisection on the logarithm of the pool level;
- one check for rising requirement tested nothing and was removed.

Code and output.

- Code: `scripts/pam_propositions_check.py`.
- Output: `scripts/pam_propositions_check_output.txt` (every check "ok"; "ALL OK").
- To run: `python scripts/pam_propositions_check.py`.
- The script's own comments use working labels P1 to P18:

Paper result	Working label
Proposition 1	P1
Proposition 2	P2
Proposition 3	P4
Proposition 4	P3
Proposition 5	P17
Proposition 6	P6
Proposition 7	P8
Proposition 8	P13

Paper result	Working label
Proposition 9	P15
Proposition 10	P9
Proposition 11	P10
S1.1	P5
S1.2	P16
S1.3	P7
S1.4	P11
S1.5	P12
S1.6	P14

S3. Prior theories against the model's components

Each row is a theory or family of models; each column is a component of the model (main text, Section 3). Table 1 of the main text condenses this table.

Codes:

- **F:** present in formal (mathematical) form.
- **W:** stated in words.
- **E:** measured, without a general model.
- **p:** partial, or a different form (see the note).
- **·:** not found in what was read.
- **?:** not checked.
- **†:** the row rests on abstracts only and is provisional.

How much of each source was read is listed after the notes.

Columns:

Column	Component
Gov	A governor that regulates access, with modes and gates
Acc	Order of loss set by access (transport, gating, affinity, pathway), per resource
Pre	That order fixed beforehand from documented access, then tested
Sto	Stores drawn first; a protected floor; intake held or raised
Uni	Units active, switched off, lost or scarred
Rep	Repair competing for access (repair cut first)
Led	A conserved ledger of load: unmet requirement relocated, not removed, including what the held record does not show
Via	Viability: collapse against death; a threshold switch before passive failure
Scr	Scars and the template (lasting residue; replacement needs a template)
Res	More than one resource
Sys	Systems other than bodies
Tst	Pre-registered tests

Theory	Gov	Acc	Pre	Sto	Uni	Rep	Led	Via	Scr	Res	Sys	Tst
Dynamic Energy Budget theory (Kooijman 2010)	p ¹	p ²	.	F	.	. ³	p ⁴	F ⁵	p ⁶	F	p ⁷	.
Selfish Brain (Peters et al. 2004; Peters and Langemann 2009; Göbel and Langemann 2011)	F	F ⁸	p ⁹	F	.	.	p ¹⁰	.	.	.	.	E ¹¹
Energetic model of allostatic load (Bobbá-Alves, Juster and Picard 2022)	W	W ¹²	.	W	.	W	W ¹³	.	W	.	p ¹⁴	.
Brain-body energy conservation model (Shaulson, Cohen and Picard 2024) [†]	W	W	.	?	.	?	?	?	W	.	.	.
Plant hierarchical models (Grossman and DeJong 1994; reviewed by Lacoine 2000 and Marcelis and Heuvelink 2007)	.	. ¹⁵	.	p	p ¹⁶	.	p ⁴	.	.	p	.	.
Plant transport-resistance models (Thornley 1972; Minchin, Thorpe and Farrar 1993; Minchin and Lacoine 2005)	.	F ¹⁷	.	p	.	.	p ⁴	.	.	p	.	.
Functional equilibrium in plants (as reviewed by Marcelis and Heuvelink 2007)	p ¹⁸	.	.	F ¹⁹	.	.	.	.	.	F	.	.
Allostasis (Sterling 2012, 2018; Schulkin and Sterling 2019)	W	. ²⁰	.	W	.	W ²¹	W ²²	.	W ²³	W	W ²⁴	.
Central governor (Noakes 2000, 2012)	W	.	.	W ²⁵	W ²⁶	.	.	W ²⁷	.	W	.	.
Selfish immune system (Straub 2014)	W ²⁸	W ²⁹	.	W ³⁰	.	W	W ³¹	W ³²	W	.	.	.
Hypoxia tolerance (Hochachka et al. 1996) [†]	W	.	.	?	W ³³	W ³⁴	.	W ³⁵	.	.	.	.
Triage theory (Ames 2006; McCann and Ames 2009, 2011)	. ³⁶	W ³⁷	p ³⁸	W ³⁹	p	W ⁴⁰	W ⁴¹	.	W	W ⁴²	.	.
Hierarchy of ATP consumers (Buttgereit and Brand 1995) [†]	. ⁴³	E/F ⁴⁴	.	.	.	E ⁴⁵	.	.	.	.	.	.
Haemodynamic models (Guyton, Coleman and Granger 1972; Hester et al. 2011; Barnes et al. 2020; Curcio et al. 2020)	F ⁴⁶	F ⁴⁷	.	F ⁴⁸	.	.	p ⁴	p ⁴⁹	.	p	.	.
Fetal brain-sparing models (Luria et al. 2012; Garcia-Canadilla et al. 2014)	p	F	.	.	.	.	.	.	.	p	.	.
Maintenance-Growth Model (Mauritsson and Jonsson 2023)	.	.	.	F ⁵³	.	F ⁵⁴	p ⁴	.	.	.	.	.
Disposable soma (Kirkwood 1977; Drenos and Kirkwood 2005) [†]	.	. ⁵⁰	.	.	.	F ⁵⁰	.	p	p	.	.	.
Perceptual control theory (Powers 1973)	F ⁵¹	.	.	.	.	.	.	.	.	.	?	.
This model	F	F	F	F	F	F	F	F	F	F	F	E⁵²

Notes

1. DEB adds control species by species; the κ rule is fixed, not governed.
2. DEB's account of κ through carrier densities, and its synthesising units, are access-like at the receiving end.
3. DEB has no repair process: damage is irreparable. Defence sits in maturity maintenance, which is "more facultative" than somatic maintenance (cut first).
4. Mass and energy (or carbon, or fluid volume) are conserved, but there is no account of where unmet requirement lands.

5. Death at a structural floor; hazard from ageing.
6. Ageing damage accumulates; replacing damaged cells needs undifferentiated cells (the model's template), stated but not modelled as a scar rule.
7. All organisms, but not organisations.
8. Insulin-gated access: insulin-dependent (GLUT4) against insulin-independent (GLUT1) uptake, in two compartments.
9. Predictions are fixed beforehand and set against a rival theory, but the order is the theory's premise, not derived from access resource by resource. Each model term is assigned one documented mechanism (Peters and Langemann 2009, Table 1), close to the model's mapping step.
10. The supply-chain model is formal and energy-conserving: unmet flow "propagates retrograde" and builds up in front of the bottleneck (fat, blood glucose). Sprengell, Kubera and Peters (2021b) read post-stroke hyperglycaemia as the brain's own pull on supply. Supplement S10.1 sets out why the related trial is not support.
11. Pre-registered systematic reviews (PROSPERO) of the theory's predictions against a rival theory (Sprengell, Kubera and Peters 2021a, 2021b).
12. "The most urgent processes divert or steal energy from less urgent ones": urgency, judged after the fact.
13. The cut to growth, maintenance and repair can occur "without elevating total energy expenditure", visible only in molecular sequelae.
14. The same process at whole-body, cellular and mitochondrial levels: nested levels, not other kinds of system.
15. Strict priority among organ groups, but the sequence is assigned, not derived from access.
16. Fruit abortion at a low source-sink ratio, the nearest thing to a loss rule.
17. "Sink priority being an emergent property of the model" (Minchin and Lacoite 2005), arising from transport resistance and sink kinetics.
18. Teleonomic: the plant behaves as if seeking a root-shoot balance.
19. Shortage of a resource shifts allocation to the organ that takes it in, as the model's intake is held.
20. "To each organ according to its need": priority set by the brain's prediction of need, not by access.
21. An anabolic (grow and repair) mode set by the circadian clock.
22. In treated hypertension, each blocked route moves the load to the next, until the cost lands on exercise (Sterling 2018).
23. Arteries thicken and stiffen, and the system loses the ability to return to normal pressure.
24. Society and the planet as regulated systems: an essay argument, not a mapping.
25. Exercise always ends with a reserve: 35 to 60% of muscle recruited.
26. Motor units are recruited and de-recruited.
27. Exercise is stopped "before there is a catastrophic failure of homeostasis": an active switch before passive failure.
28. Two co-equal regulators, brain and immune system, each able to take control and inhibit the other. In this model they are two modes of one governor function, selected by signals; which mode wins in conflict is an open question.
29. Insulin dependence decides which organs gain from insulin resistance.

30. A non-negotiable floor (about 8,500 kJ a day) and a negotiable remainder; stores last 19 to 43 days.
31. Energy arithmetic of the reallocation (kJ a day), not a conserved account of unmet requirement.
32. Acute programmes must end within the store's time; chronic activation is "a misguided acute program".
33. Channel arrest, spike arrest and translational arrest.
34. Protein synthesis is cut (translational arrest).
35. Arrest is reversible in tolerant cells and "irreversible" in sensitive cells.
36. No central regulator: the mechanism is distributed (binding affinities, isozymes, distribution).
37. Binding affinity, transporters, preferential distribution, and a selenium-sensitive tRNA.
38. Essentiality is classified first, from knockout lethality, then checked against the order of loss (vitamin K-dependent proteins, selenoproteins). The classification is not by access and was not pre-registered.
39. Transport proteins are induced first ("homeostatic adjustments"), then triage follows.
40. DNA-repair enzymes lose to enzymes of ATP synthesis.
41. "Insidious changes accumulate" while critical functions stay intact: the cost is hidden from the held record.
42. About 40 micronutrients; worked through for vitamin K, selenium and iron.
43. Control is "widely shared"; no block has more than a third of it.
44. A measured order (macromolecule synthesis, then sodium cycling, then calcium cycling, then proton leak), analysed by metabolic control analysis.
45. Protein and RNA/DNA synthesis are the most sensitive to supply.
46. Baroreflex and other control loops in formal models.
47. The regional order emerges from bed resistances, reflex gains and autoregulation, calibrated to data.
48. Venous unstressed volume is mobilised as a store.
49. Time to cardiovascular collapse.
50. Repair investment is set by optimisation (fitness), not by access.
51. Hierarchical control: higher levels set the reference levels of lower ones.
52. PT1 (main text, Section 5.1): pre-registered and adjudicated blind; one supported result, at half weight.
53. Non-negotiable basal maintenance and feeding costs are paid first; negotiable maintenance and growth share the rest.
54. Negotiable maintenance (defence: immune system, buffering) is cut as a power of relative food intake.

What the table shows.

- **Every column except Pre, Led and Sys is filled by some prior theory**, most often in words. The model's governor, access-based order, stores, units, repair cut first and scars each appear elsewhere.
- **No prior theory fills more than about half of its row.** The nearest are:
 - the Selfish Brain: formal and tested, but one resource and two compartments;
 - the selfish immune system: in words, energy only, two regulators;

- triage theory: in words, many resources, several levels.
- **Three columns are filled by no prior theory:**
 - **Pre:** a rank fixed beforehand from documented access and then tested. Triage theory (by essentiality, not access) and the Selfish Brain (as a premise, not derived) come closest.
 - **Led:** a conserved account of where unmet requirement lands. It is stated in words by the energetic model of allostasis, by allostasis and by the selfish immune system, but never kept as an account.
 - **Sys:** the same rules applied outside bodies.

How much of each source was read:

- **In full:**
 - Bobba-Alves, Juster and Picard 2022; Peters and Langemann 2009;
 - Sprengell, Kubera and Peters 2021a and 2021b; Sterling 2018;
 - Noakes 2012; Straub 2014; Ames 2006; Mauritsson and Jonsson 2023;
 - Marcelis and Heuvelink 2007 (for the plant model classes).
- **Sections read:** Kooijman 2010.
- **Abstracts only:**
 - Peters et al. 2004; Göbel and Langemann 2011; Shaulson, Cohen and Picard 2024;
 - Lacoïnte 2000; Minchin and Lacoïnte 2005;
 - Sterling 2012; Schulkin and Sterling 2019; Noakes 2000;
 - Hochachka et al. 1996; McCann and Ames 2009 and 2011; Buttgerit and Brand 1995;
 - Barnes et al. 2020; Curcio et al. 2020; Garcia-Canadilla et al. 2014; Luria et al. 2012;
 - Drenos and Kirkwood 2005; Powers 1973.
- **Cited for the tradition they founded, not read here:**
 - Thornley 1972; Minchin, Thorpe and Farrar 1993 (title and review accounts);
 - Grossman and DeJong 1994 (via Marcelis and Heuvelink 2007);
 - Guyton, Coleman and Granger 1972; Hester et al. 2011;
 - Kirkwood 1977.

S4. All predictions, with status

The model's predictions are G1 to G26, in their current wording (model v0.19, Section 10, in this record).

Layers: layer 1 holds for any feedback loop with a finite store; layer 2 needs selection or design.

Evidence labels:

- **derived:** follows from the reduced form; the proposition is named;
- **simulation result:** reproduced by the simulation engine used during the model's development (the engine is not part of this record);
- **natural-system observation:** consistent with published findings read for the checks in S6, which are not tests;
- **compatible:** the direction fits, but the finding cannot discriminate;
- **direct test:** a pre-registered held-out test (main text, Section 5.1);

- **not yet tested.**

Numbering gaps:

- **G11 was withdrawn** during development: it followed from existing rules and added nothing.
- **G9, G14, G15 and G17 are excluded from this paper.** They were carried from an earlier version without re-examination, so they are listed below as excluded, not as predictions.

No.	Prediction	Layer	Status
G1	The record stays near normal while lower parts and stores move. It is flat while the gap is no larger than what the stores release plus what parts below the top can give up	1	Derived (Proposition 2); simulation result; natural-system observation: consistent. The H1 check was not consistent (10.7% of 75 cases), at low weight
G2	Lower-ranked parts lose access first, in ascending rank	2	Derived (Proposition 4); simulation result; natural-system observation: consistent
G3	With a store carrying the gap and its release not binding, the break comes at the same cumulative shortfall whatever the rate. Where release binds (a tapering store, or a gap above the release rate), faster onset breaks earlier. Under proportional release, the store left unused at the break is (gap minus margin) divided by the store's turnover rate, so a faster shortfall leaves more of the store unused, in proportion	1	Derived (Proposition 3). Not yet tested; fast and slow haemorrhage in sheep and fast and slow drought in trees fit the first clause (main text, Table 4)
G4	A larger store gives a longer silence; a depleted start breaks sooner	1	Derived (Proposition 2); natural-system observation: consistent
G5	Parts absorbing load lose units (switched off first, lost if supply falls too fast), so loss compounds down the ranks	1	Derived (Propositions 4 and 6); simulation result
G6	Recovery starts with the intake; the record recovers before the state does	2	Simulation result; natural-system observation: consistent (kidney, S6)
G7	Fixed capital keeps what it loses; renewable parts rebuild lost units, except where the template is scarred	1	Simulation result; natural-system observation: consistent
G8	Rate decides harm. Renewal falling no faster than a part can switch units off leaves no loss. Faster falls lose units. A scar needs speed and depth together; the threshold depth falls as speed rises	1	Derived with bounds (Proposition 6); simulation result. Not yet tested
G10	After a short episode, parts come back before stores; after a long one, stores come first	2	Simulation result; natural-system observation: consistent. Qualitative unless the costs are mapped (Proposition 10)
G12	Recovery from small knocks slows before the break when the break is approached through shrinking release headroom (a tapering store, or a gap rising towards the release rate). It does not when the break comes with headroom intact (a store with full release emptying, or a switch)	1	Direct test: failed in H1 (blood loss under anaesthesia; release profile committed as a taper), at half weight. Runnable only under the pre-data reading of missing bins; the step-down inside the fallback cohort was not stated in the pre-registration. Lead time derived (S1.1)
G13	Lowest-ranked parts take the load first, and longest where their value in recovery is also lowest (recovery is by marginal value, not rank)	2	Derived (S1.5); simulation result
G16	Economising: with an expected shortfall, the governor cuts work access early and evenly. It causes no unit loss at any speed, because work is cut, not renewal. Its saving arrives at once for work and wear, and as units are switched off for their baseline renewal; stores are preserved	2	Derived (S1.4); simulation result

No.	Prediction	Layer	Status
G18	Parts sharing a loaded dependency move together before the break; parts clustered only by rank do not	1 and 2	Simulation result. Not yet tested
G19	Three outcomes (switched off, lost, scarred) can be told apart by what comes back and how fast	1	Simulation result. Not yet tested
G20	Exhaustion and severance. No part loses units while there is somewhere else to take resources from, and the top goes last. Failure with willing receivers intact means a link cut or constricted below need (pathway capacity as maximum flow), or damage from outside to the top or its supports	1	Derived (Proposition 7); simulation result. Not yet tested
G21	The law of the minimum. A part's work falls in proportion to its scarcest resource while other resources are ample; what it cannot use stays in the flow, and surplus is spilled	1	Simulation result. Known (Liebig's law); general form in DEB's synthesising units (Kooijman 2010)
G22	The intake coasts. With nothing to take in, the intake switches its units off on basal maintenance and is back first at refeeding, whatever its rank	2	Derived (Proposition 10); simulation result
G23	Repair is a shared, governed network. (a) Under a sustained shortfall, repair slows across the system, even where the damaged part's own supply is ample. (b) Strict rank: when several parts are damaged at once, those ranked after another damaged part heal more slowly than alone; the highest-ranked damaged part heals as fast as alone. (c) An acute threat moves repair capacity to likely sites of damage before it occurs	2	(a) Compatible: psychological stress slowed wound healing (Kiecolt-Glaser et al. 1995; Marucha, Kiecolt-Glaser and Favagehi 1998). (b) Derived (Proposition 11); not yet tested; it can be told apart from shared repair, in which every damaged part slows. (c) Compatible: acute stress hormones enhance skin immune function (Dhabhar and McEwen 1999)
G24	Cascade along substitutes. Losing or narrowing a route moves its flow onto the rest. Where flow divides by physics (vessels, pipes, power lines), a surviving route can be overloaded, or the displaced flow drawn from another part's branch (steal), even with spare total capacity. Where flow is reallocated by choice (budgets, routers), this happens only when total spare capacity is short. Overloaded routes fail in turn, so failure spreads along the substitutes, not by rank	1	Derived (Proposition 8). Known in power systems, and steal is known in physiology; not yet tested as a cross-domain prediction
G25	Order of loss from access. Under scarcity, the order in which parts lose adequate supply is predicted by properties of access documented beforehand (constriction under sympathetic drive, autoregulation, affinity, redundancy; discretionary budgets)	2	Direct test: supported in PT1 (English single-tier councils, 2014-15 to 2019-20, statutory duty classified blind), at half weight: lines with a statutory duty were protected more than discretionary ones. Lines with a duty of uncertain level were not distinguishable from discretionary ones. The scarcity version (G25-C), that the gap widens where funding fell more, was not supported. Compatible: splanchnic and renal constriction in haemorrhage (main text, Table 4)
G26	Partial refill. In recovery, a store refills only until its marginal value falls to that of the best competing use, so it need not refill to full while parts are still short. The refill level is the critical fractile $F^{-1}(1 - V/(c_S\rho))$ of remembered episode depths	2	Derived (Proposition 10). Not yet tested

Excluded from this paper. These are carried from an earlier version and were not re-examined:

- G9, re-tuning to a past threat;
- G14, peak-referenced protection;
- G15, store memory after a deeper shortfall;
- G17, economising and growth compete.

S5. Held-out test records

Both tests followed the same procedure, in this order:

1. Documentation was gathered by another model, barred from reporting values.
2. A setting map and mapping were written.
3. The pre-registration was frozen.
4. The analysis code was written and tested on synthetic data only.
5. The data were downloaded and counted, then run once with the code unchanged.
6. The computation was replicated by a second AI system, which wrote its own script without seeing the data or the results.
7. A blind adjudication was made against the frozen tables and the adjudication rules (`tests/ADJUDICATION_RULES.md`), in a separate session with no shared context.

Every unplanned choice is logged as a deviation, with the direction it pushes. Fingerprints (SHA-256) of the frozen files are in the repository README.

S5.1 H1: warning before decompensation in surgical blood loss (G12)

Item	Record
Documentation	raw/2026-10-06_gemini_H1-VDB-S1.md (structure only; no values)
Setting map and mapping	tests/H1 VitalDB G12 - setting map and mapping (draft).md
Pre-registration	tests/H1 VitalDB G12 - pre-registration.md, frozen 6 October 2026 (SHA-256 begins 371caf3c)
Model version tested	theory/TIER_QUEUE_MODEL_v0.17.md
Analysis code	tests/scripts/h1_vdb_g12.py (SHA-256 9fbe80d7), unchanged since before the data were opened
Synthetic checks	tests/results/H1-VDB-synthetic/: 7 of 7 verdicts as built. G1 was not challenged by any synthetic case
Count and result	tests/results/H1-VDB/ (counts.json, summary.json, README.md)
Replication	raw/2026-10-06_chatgpt_H1-VDB-R1.md; tests/results/H1-VDB-R1/
Adjudication	raw/2026-10-07_chatgpt_H1-VDB-ADJ1.md

Data and cohorts. VitalDB (Lee et al. 2022).

- **Primary cohort (no fluid boluses):** 755 eligible cases, with no qualifying high-loss case, so it was not runnable.
- **Fallback cohort:** 2,324 eligible cases. At a loss threshold of 20% of estimated blood volume the high-loss group was short (19 cases). At 15% it gave 25 high-loss and 42 low-loss cases, analysed at half weight as fixed in advance.

Result.

- **P1, the excess rise in lag-1 autocorrelation of mean arterial pressure before falls in high-loss cases:** median -0.072 , one-sided $p = 0.48$.
- **P2, high-loss excess greater than low-loss:** $p = 0.78$.
- **Verdict as computed:** Fails, and not Contradicted (two-sided $p = 0.96$).
- **Sensitivity analyses** (not part of the verdict): Fails in every computable variant.

- **G1 check** (low weight): pressure held while haemoglobin fell in 10.7% of 75 cases, against a pre-stated "more than half". Not consistent.
- **G18** (exploratory, not scored): the correlation of pressure with heart rate rose by a median of 0.126 more before falls than across controls (25 cases).

Replication. The second system's script reproduced the frozen run exactly under the frozen reading of missing 10-second bins, with every count, score and test identical. Under its own stricter reading (a missing bin fails the stable approach), the test is not runnable (15 high-loss and 31 low-loss cases). The frozen run was kept as the result of record, with this dependency stated.

Adjudication (verbatim): "VERDICT FOR G12 IN THIS SYSTEM: Fails." Weight: "fallback cohort, half weight; 15% EBV high-loss threshold." The adjudicator recorded two qualifications that go with the verdict:

- the test is runnable only under the pre-data reading of missing bins;
- the pre-registration does not state that the step-down to 15% applies inside the fallback cohort, although the code fixed before data applies it.

Deviations: none from the pre-registration or the code.

S5.2 PT1: fixed against dynamic priority, and order of loss from access, in English local government (G25)

Item	Record
Documentation	raw/2026-10-07_gemini_PT1-S1.md, raw/2026-10-07_gemini_PT1-S2.md
Blind classification of statutory duty	raw/2026-10-07_gemini_PT1-C1.md (failed, without files), raw/2026-10-07_chatgpt_PT1-C1b.md, raw/2026-10-07_chatgpt_PT1-C1c.md; frozen as tests/PT1_statutory_classification.csv (176 lines; SHA-256 begins faf45875), with tests/PT1_line_name_crosswalk.csv
Design note, setting map and mapping	tests/PT1 Priority test - design note (draft).md; tests/PT1 Priority test - setting map and mapping (draft).md
Pre-registration	tests/PT1 Priority test - pre-registration.md, frozen 7 October 2026 (SHA-256 begins a4aab81d)
Model version tested	theory/PERSISTENCE_ALLOCATION_MODEL_v0.18.md, as corrected on 7 October 2026
Analysis code	tests/scripts/pt1_councils.py (SHA-256 995abfc5), unchanged throughout; parser tests/scripts/pt1_parse.py
Synthetic checks	tests/results/PT1-synthetic/: 6 of 7 scenarios as built. The seventh was a chance false positive and was correct on reseed
Procedure log and deviations	tests/PT1 Priority test - procedure log.md
Result	tests/results/PT1/ (counts.json, summary.json, run.log, README.md)
Replication	raw/2026-10-07_chatgpt_PT1-R1.md; tests/results/PT1-R1/
Adjudication	raw/2026-10-07_chatgpt_PT1-ADJ1.md

Data.

- **Sample:** 121 single-tier councils in England, 2014-15 to 2019-20; 93 spending lines; 23,373 rows for PT1 and 18,905 for G25.
- **Access property:** whether a statutory duty attaches to each line, classified blind before any spending data were opened:

- A: a duty;
- B: a duty of uncertain level;
- C: discretionary.

Deviations, each with its direction:

- **D-1:** a name crosswalk for 21 lines whose names changed between years. Decided from names only, before values were seen. Neutral.
- **D-2:** many-to-one lines summed. Neutral.
- **D-3:** a labelling error in the source relabelled by position. Neutral.
- **D-4:** a limitation, with no intervention: allotments are reported separately from 2019-20.
- **D-5:** duplicate codes removed from the funding table after the first run stopped with an error before producing any result. Approved by the author. Neutral. The analysis code was unchanged.

Result.

- **PT1, primary:** $\beta = 0.054$ (SE 0.405), one-sided $p = 0.45$, 95% interval -0.75 to 0.86 .
Inconclusive: the interval includes the smallest effect set in advance as meaningful (0.25).
- **G25 (half weight):** within a council and year, lines with a statutory duty grew in real spending per head about 4.6 percentage points a year faster than discretionary lines (one-sided $p = 2 \times 10^{-11}$). Class B was not distinguishable from C. **Supported.**
- **G25-C (half weight):** the interaction with the depth of the funding fall was 0.22 (SE 0.25). **Not supported.**
- **Sensitivity analyses** (not part of the verdicts):
 - PT1 is inconclusive in every variant;
 - G25's protection of class A holds in every variant;
 - the B-above-C step fails without London.

Replication. The second system's script reproduced every count, estimate, standard error, p-value and verdict to at least eight significant figures. One descriptive (unscored) count differed because of a definitional choice.

Adjudication (verbatim): "VERDICTS: PT1 Inconclusive (full weight); G25 Supported (half weight); G25-C Not supported (half weight)." No faults were found that make a verdict unsafe. None of the adjudication rules changed a verdict.

Contamination, declared in the pre-registration: G25 carries half weight because the AI collaborator knew the broad pattern in advance.

S6. Natural-system observations

What these are. Twelve surface-level checks, made on 5 October 2026 while the model was being built.

- **Method:** for each, predictions were written down and committed to the repository before any source was opened. Published findings were then read and each prediction was given a verdict.
- **These are not tests,** for three reasons:
 - many outcomes were known in advance to the AI collaborator, and each check declares this;
 - sources were mostly read through search summaries;

- three early refinements of the model were built from the blood-loss and fasting findings.
- **Weight:** they show that the model is compatible with these systems, nothing more. They carry no weight in the paper's claims.
- **Full records:** the prediction tables, findings with their sources as read, contamination statements and weights are in `theory/natural_test_*.md`, with the predictions committed in earlier commits than the results.
- **Sourcing:** findings that the main text relies on are cited there from sources read directly (main text, Table 4). Where a later reading corrected a summary-based finding, the correction is noted below.

No.	System	Predictions	Verdicts	Contamination and weight
1	Blood loss and lower-body negative pressure	N1 to N6	3 consistent; N2 partly (lower-priority beds cut first, but brain supply only partly protected); N3 steepness consistent, mechanism different; N6 partly	Class scheme and compensatory reserve known; N4 rests on 8 sheep (Scully et al. 2016)
2	Prolonged fasting	F1 to F7	4 consistent; F6 consistent for starvation but the opposite under food restriction (weak; summary only); F4 and F7 not tested	F1 and F3 known; summaries only, including the threshold source (Cherel and Groscolas)
3	Loss of kidney filtering units	K1 to K6	6 consistent	Summaries only; K1 and K3 known
4	Honeybee colony under worker loss	H1 to H7	5 consistent; H6 and H7 partly	Heavy: most known in advance; H4 at the level of individual bees
5	Plants in drought	P1 to P8	6 consistent; P6 mostly, with one finding against (S7); P8 not tested	P2, P3 and P5 known. P2 (leaves fail first) was later found contested (Peters and Choat 2025; S7). A figure for embolism in green canopies was not found in its cited source and is withdrawn
6	Fetal growth restriction	FG1 to FG7	7 consistent; FG5 favoured one of two recovery rules	FG3, FG4 and FG6 known; FG5 contaminated
7	What sets the recovery order (two rival rules)	Three discriminating cases	Rule L (type of past load) fitted all three; Rule D (next demand) fitted none as the main driver	Case A contaminated
8	Muscle: acute injury, chronic overload, disuse	M1 to M8	6 consistent; M3 partly (the record was not clearly late in muscle); M4 not tested	Mostly known
9	What counts as a record (five paired cases)	R1 to R5	5 consistent	Heavy: all five known in outline; R2's human evidence contested
10	Re-tuning after past threats	RT1 to RT6	5 consistent (RT5 on benefit only); RT6 partly (holds in one group only)	Mostly known in outline
11	Protection referenced to peak load, in muscle	FM1 to FM5	2 consistent; FM2 and FM5 partly; FM3 inconsistent as written	Summaries only; FM1 and FM4 known
12	Weight cycling	WC1 to WC7	2 consistent; 3 partly; WC6 and WC7 not found	Partly known

Checks 6, 10, 11 and 12 bear partly on predictions excluded from this paper (G9, G14, G15; S4).

What recurs across the first five systems (corrected where later reading required):

- A defended figure is held steady while a buffer is drawn down:
 - blood pressure in blood loss;
 - the function of protected organs in fasting;
 - creatinine in kidney loss;
 - visible brood and stores in a colony losing workers;
 - canopy colour in drought.
- **State markers show the drawdown and the figure does not:** stroke volume and compensatory reserve; fat mass; renal functional reserve; adult numbers; loss of hydraulic conductivity.
- **The break comes at a set depletion, not at exhaustion, and often through an active switch:**
 - sympathetic withdrawal at about 30% blood loss;
 - phase III of fasting at a threshold fat share;
 - creatinine rising after about half of the filtering units are lost;
 - brood cannibalism when protein runs short;
 - about 80% loss of conductivity in loblolly pine saplings.
- **The starting state sets how much room there is:** heat stress shortened tolerance to blood loss; a leaner start reached phase III sooner; a low starting number of filtering units led to earlier decline.

Two cases moved here from an earlier draft of the main text (illustrations only):

- **Kidney.** Creatinine stays roughly normal until about half of the filtering units are lost; the remaining units filter more, and the renal functional reserve falls first. After acute injury, creatinine returns to baseline while reserve does not. This is from search summaries; K1 was known in advance.
- **Fasting penguins and other birds.** Phase III of a long fast, with protein breakdown, renewed food search and, in breeding birds, egg desertion, begins at a threshold fat share (about 9% in penguins) rather than after a set time. A larger starting store lengthens phase II (Cherel and Groscolas). The model's point at which a store begins to release was built from this source, so this case is a basis, not a check.

Other sources are listed, as read, in each check's file.

S7. Findings logged against the model

Rule. Every finding that does not conform, or may not conform, is logged when it is found, with its source.

- **Weight:** it is weighed by the same standards of data quality and reporting as a supporting finding.
- **Explanations:** an explanation may be offered only if it names the specific place the load landed, or the specific fault in the measure or the mapping. It is logged as a prediction before it is checked, and the outcome is logged either way. If the named place or fault is not found, the explanation is struck.

- **Absence:** "not identified" is not "absent". A null result on the measures a study took is weighed as a result about those measures only.

S7.1 Held-out tests

Finding	Bears on	Weight	Explanation logged in advance
G12 failed in H1: no rise in lag-1 autocorrelation of arterial pressure before falls in high-loss surgical cases (S5.1)	G12	Half weight (fallback cohort); runnable only under the pre-data reading of missing bins	None checked yet. The first candidate is the mapping (the store's release profile under anaesthesia, where an anaesthetist acts as an outside loop holding pressure). It must be named in advance of any new test
G1 check in H1: pressure held while haemoglobin fell in 10.7% of 75 cases, against "more than half"	G1	Low (a check, not a test)	None
G25-C not supported in PT1: the protection gap did not widen where funding fell more (S5.2)	G25-C	Half weight	None
PT1's class B (a duty of uncertain level) was not distinguishable from discretionary lines, and the B-above-C order fails without London	G25	Sensitivity only	None

S7.2 Literature

Finding	Source	Bears on	Weight and status
After prolonged starvation, heart, liver, pancreas and kidney each lost about 40% of their mass while the brain lost under 2%, with no order among the organs below the brain	Krieger (1921), as cited by Peters and Langemann (2009); original not read	G2 (order of loss below the top)	Not yet weighed. Whether it conflicts depends on the order fixed for energy and protein at a starvation mapping
In 12 Australian tree species, the predicted leaves-before-stems order of hydraulic failure (vulnerability segmentation) was "universally absent or negative"	Peters and Choat (2025), abstract	Natural check 5, prediction P2; not used in the main text	Logged against P2; the leaves-first order is not claimed
In mature spruce, an earlier drought eased physiological stress in a later one, where a stressed start should fare worse	Hikino et al. (2026), abstract	G4	Explanation named in advance and checked; it holds. The earlier drought cut leaf area by 60%, still 30% four years later, so demand on the same water supply was lower. Recorded as a scar that reduces demand, not counted against G4
Human food insecurity is linked to higher body weight only in adult women in high-income countries	Nettle, Andrews and Bateson (2017), as summarised	Natural check 10 (RT6)	Logged; the model offers no reason for the restriction
Synaptic loss after noise is shown in animals, but one human study with normal audiograms found no evidence of it	Guest et al. (2018), as summarised	Natural check 9 (R2)	Logged as a caveat on that case

S7.3 Natural checks (S6)

Blood loss, mismatch 1: brain blood flow fell by 27% at presyncope while arterial pressure was unchanged; the top was not fully protected before the break.

- **Explanation named at the time:** the brain's supply depends on cardiac output, which was itself falling.

- **Outcome:** the model was revised so that the top can share an upstream supply.
- This mismatch was used to build the model, so it no longer counts as a check.

Blood loss, mismatch 2: decompensation is an active switch, not a passive failure of control. The model was revised to include a threshold switch, built from this finding.

Fasting (F6): recovery order was consistent for starvation but the opposite under food restriction. Weak, because only a summary was read.

Muscle (M3): performance (the record) fell alongside mood in overreaching, so the record was not clearly late.

- **Explanation named:** maximal performance tests probe capacity directly, so they are state signals.
- **Outcome:** check 9 tested that rule on five cases, all consistent.

Protection referenced to peak load (FM3): inconsistent as written. A reassignment was made after the result and checked once; it is weak.

Recovery order: the rival rule D (dependency and next demand), named in advance, did not hold as the main driver and was struck in that role.

Weight cycling: two results against are logged; neither has been read beyond its summary:

- no slower loss in the second cycle in female rats (Archambault et al. 1989);
- a rodent study summarised as finding no adverse effect of cycling.

S7.4 A corrected entry, kept for the record

The finding as first logged: in tumour-bearing mice, the brain's glucose use fell while the tumour became the second-largest glucose consumer. It was logged as against brain priority.

The correction: the full abstract states that "brain metabolism in the tumour-bearing state was maintained by an increased use of lactate and 3-hydroxybutyrate" (Mulligan and Tisdale 1991). The brain switched fuel; its energy supply was not cut. For the brain the resource is energy, so the finding is not against brain priority.

The entry is kept to show the correction.

S8. Literature searches

Three searches looked for formal models that set an order of loss among parts under shortage. Each protocol was committed before the search ran. Strings, scripts, raw records, screening sheets (a decision and reason for every record) and counts are in `theory/search_check1/`, `theory/search_check1_s2/` (with `theory/search_check1_s2_run1/`) and `theory/search_check2/`. Screening was by title and abstract, by one AI reviewer, except where stated.

S8.1 Check 1, search 1

Protocol: `theory/PAM_check1_search_protocol.md`, committed before running.

Question: has any published formal model specified the order in which two or more organs, tissues or body parts lose supply under resource shortage, and is that order derived from properties of access?

Databases and strands: PubMed and OpenAlex, all years, run on 7 October 2026. Strand A covered dynamic energy budget models of organs under shortage; strand B covered any formal model of organ priority under shortage.

Classes for includes:

- I-access: order from properties of access;
- I-fixed: order assigned;
- I-emergent: order from another mechanism.

Formal priority among functions (F) and empirical orders with no model (E) were recorded separately.

Counts:

Step	Number
Retrieved	150 records
After deduplication	139
Included	0
Formal function priority (F)	4, all dynamic energy budget models
Empirical order, no model (E)	2

Result: not usable. The search failed its own known-item check. It did not retrieve the one relevant formal model already known (Göbel and colleagues' brain-against-periphery model), and its terms missed plant source-sink allocation. Its null result is reported and not used.

S8.2 Check 1, search 2

Protocol: theory/PAM_check1_search2_protocol.md, committed before running. It was built to close search 1's two gaps:

- **strand C:** plant source-sink allocation models;
- **strand D:** organs competing for energy.

Sensitivity test: set in advance as six known items.

- **Design items (could be used in revision):**
 - K1: Göbel and Langemann 2011 (published online 2010), *Theory Biosci*;
 - K2: Peters and Langemann 2009;
 - K3: Minchin, Thorpe and Farrar 1993;
 - K4: van Leeuwen, Zonneveld and Kooijman 2003.
- **Held out (could not be used in revision):**
 - H1: Thornley 1972;
 - H2: Göbel et al. 2010, *J Theor Biol*.

Revision 1, the only one allowed. Run 1 missed K4, which uses the vocabulary of energetics ("tumour-in-host", "energetics of tumour and host"), not of allocation. A fourth sub-query was added to strand D, and strand C was unchanged. Run 1's records are kept.

Known items, final: K1 to K4 found; H2 found; **H1 missed.**

- The miss is explained by vocabulary alone. H1 is not in PubMed, and neither its title nor its OpenAlex abstract contains any strand C phrase: it speaks of root:shoot ratios, not allocation.
- **Strand C is therefore of limited sensitivity**, as the protocol requires it to be reported.

Deviation D-1 (approved by the author): strand C screened through reviews. Strand C's exhaustive title pass (11,175 titles) was stopped after 410, because the question it was built for was already answered: formal allocation models among plant organs are common, and at least one (K3) derives sink priority from transport properties.

- **Replacement:** a targeted reading of reviews of plant allocation models, chosen by a title rule fixed in the deviation. The reviews are Lacoite 2000, Génard et al. 2008, Franklin et al. 2012, Minchin and Lacoite 2005, and Marcelis and Heuvelink 2007, the last read in full.
- **Result:** four classes of plant model (S3):
 - functional equilibrium;
 - sink regulation by strict priority (hierarchical models);
 - transport-resistance models, in which priority emerges;
 - optimisation models.

Deviation D-2 (a method decision, logged): the OpenAlex title pass of strand D done by AI agents, with a blind check.

- **The task:** 6,809 new titles needed a title pass. Three AI agents in sequence did it under a deliberately liberal rule: keep anything that could be a formal allocation model or a neighbouring theory.
- **The check:** before any agent ran, the reviewer screened a random sample of 300 titles blind (seed 20261007). The agents kept all 10 of the reviewer's sample keeps and marginal cases, plus 12 more.

Counts:

Step	Number
Strand C retrieved	PubMed 1,616; OpenAlex 11,834
Strand D retrieved	PubMed 385; OpenAlex 7,372 (6,809 new)
Final deduplicated set	18,363 records

Includes (strand D, both databases): 21.

- **12 I-access:**
 - the Selfish Brain formal models (Göbel and colleagues 2010, 2011, 2013; the 2012 review of mathematical models; the 2008 deductive appetite model; the supply-chain models of 2007 to 2011);
 - a tilt model of cerebral perfusion;
 - a brain-centred four-compartment model (2026 preprint);
 - a lifespan brain-health model built on Göbel et al. 2010.
- **7 I-emergent:** dynamic energy budget tumour-in-host models (van Leeuwen et al. 2003; Tosca and colleagues, 2018 to 2021); and a spreading-depression model.
- **1 weak I-emergent:** a GLUT1-deficiency model.
- **1 weak I-fixed:** a 1982 computer model in which an autonomous tumour drain draws first.

Not included:

- 1 provisional I-access (abstract only);
- 5 records still to check (title or abstract only), none likely to change the answer;
- 1 probable exclude;
- 18 near misses (dynamic energy budget models of two organisms: host and symbiont, host and parasite);
- 55 neighbouring theories (to S3);
- 33 empirical records;
- 19 formal models with no allocation under shortage.

Answer to check 1. Formal models with an order of loss among parts exist in three domains: plants (strict priority, or priority emerging from transport), human energy metabolism (the Selfish Brain family, with access gated by insulin), and tumour-host energetics (dynamic energy budget). None has:

- unit loss or scars;
- repair competing for access;
- a ledger of unmet requirement;
- viability;
- use across domains.

S8.3 Check 2: models of the order of organ loss in starvation and haemorrhage

Search: PubMed only, run on 7 October 2026. Five strings, with one revision logged before screening (theory/search_check2/README.md). It returned 1,279 unique records, all titles read, and candidates read as abstracts.

Answer:

- **Haemorrhage: a large formal tradition exists.** It covers the Guyton model and its descendants, and lumped-parameter closed-loop models with baroreflex control (S3). The order among vascular beds is not written in as a rule. It emerges from each bed's resistance, reflex gain and autoregulation, fitted to data: order from properties of access.
- **Fetal brain sparing:** formal circulation models, with the order again set by bed resistances.
- **Starvation across many organs:** no formal model of the order of organ loss was found. The formal models found have two compartments (fat and lean; brain and body). The multi-organ order is held as data, not as a model.

Limits: PubMed only; one reviewer; abstracts only. "Not identified" is not "absent".

S9. Mapping a system, in full

The model makes predictions about a system only once the system is mapped. Every choice below is fixed before any outcome data are opened. Anything chosen with the expected outcome in mind is declared fitted. The two held-out tests show worked mappings:

- tests/H1 VitalDB G12 - setting map and mapping (draft).md;
- tests/PT1 Priority test - setting map and mapping (draft).md.

S9.1 The nine steps

1. **Boundary, currency and protected level.** Name the system, its boundary and the level whose viability is protected. These three are the model's main safeguards.
 - **Viable set:** fix it, and fix whether outside support counts as admissible, since that decides collapse against death (Proposition 9).
 - **Records in a pathway:** a record may be an early stage of a longer pathway, which can hold while load appears in a later, dependent stage. Fix whether viability requires the early output, the downstream completion, or both. The order of failure in what is observed need not be the order of onset.
2. **Resources, carriers and stores.**
 - Which resources does work need, and in what ratio?
 - Are they complementary (used in fixed proportion)? The law of the minimum applies only where they are. Substitutable inputs are mapped as one resource, or declared (for the brain, glucose and ketones are one resource, energy).
 - What carries them (the network)?
 - Which stores exist for each, in what order, with what release profile (proportional, or full until empty) and what turnover?
 - Which can be spilled?
3. **The governor and its levels.** What is the record, and which levels does it depend on? What does the governor act through: stores, intake, gates, pathway capacity, explicit budgets?
4. **Parts.** For each part, fix the following from independent sources:
 - its units and capacity;
 - its rank for each resource, fixed from documented properties of access, or one shared rank if the orders are documented to coincide;
 - where access is by saturable uptake, its half-saturation constant, maximum uptake and reference allocation (S9.3);
 - its rebuild time, template limit and route back (for fixed capital, none);
 - its requirements per unit of work, basal maintenance and renewal (a baseline per active unit plus wear per unit of work);
 - its reference allocation, against which load is counted;
 - whether it is support for the top, whether it is the intake, and whether it is temporary;
 - how economising is realised in it: consolidation (units switched off; the default), or a declared share of units working below capacity.
5. **The repair network:** its parts (resident and mobile), their capacity and source, their resources and reserves, and their rank.
6. **Links and pathways:**
 - dependencies (supply edges) and loops;
 - pathways and their routes (redundancy), and whether flow among routes divides by physics or is reallocated by choice (Proposition 8);
 - non-bypassable links (points of severance);
 - which parts share the repair network.

7. **Nested systems:** only if the proposed extension is used. List every component treated as a nested system, with the evidence for its own access loop, and its interfaces. A component not listed stays a part.
8. **Modes and gates:** only if access-limited load is claimed (S9.2).
9. **The clock:** fix the unit in which durations are judged.

S9.2 The access guard

Load has three origins:

- falling supply;
- rising requirement;
- a governor setting (a mode) that restricts access while resources suffice.

The third is admitted only for a mode and a gate documented independently, and named at mapping before outcomes. A gate not named at mapping cannot be invoked later to explain a shortfall. This keeps the access-limited regime of Proposition 1 from becoming a way to explain away any result.

S9.3 Rank under saturable uptake

Where parts take a shared resource by saturable uptake (transporters, binding), document for each part:

- the half-saturation constant K ;
- the maximum uptake V ;
- the reference allocation q^0 .

How the rank follows:

- A part loses adequate supply when the pool falls below $C^* = K r / (1 - r)$, with $r = q^0 / V$.
- **The rank is the order of C^* , highest first** (Proposition 5).
- Affinity alone decides the rank only where r is equal across parts.
- Priority is strict only where neighbouring thresholds are far apart, and shared otherwise.

S9.4 The bias ledger

Before anything is looked up, each mapping choice is scored for its interpretive freedom, from 1 (forced) to 5 (open). The choices scored are:

- the boundary;
- part classification;
- the resource;
- rank;
- the reference state;
- the prediction.

Coin-flip choices are marked as ambiguous before the search, and cannot be resolved afterwards in the model's favour.

S9.5 Using the model in reverse

Reverse mode starts from what is seen failing. It points to where load entered, which parts were absorbing it unseen, and which dependency the failing parts share.

- **It suggests where to look; it never names a cause.**
- **Disciplines:**
 - it generates hypotheses and is not evidence;
 - it corrects for what is visible;
 - it names a set, not a cause.
- It must tell failure clustered by rank from failure clustered by dependency (G18).

S10. Compatible illustrations (not tests)

Cases whose direction fits a proposition but which cannot bear weight. They are known outcomes, read after the model was built.

S10.1 Intensive insulin after ischaemic stroke (Proposition 1, scarcity regime).

- **The trial:** Rosso et al. (2012), INSULINFARCT, *Stroke* 43:2343-2349 (abstract read; closed access). 180 patients with hyperacute ischaemic stroke were randomised to intensive or usual subcutaneous insulin for 24 hours.
- **Results:**
 - intensive insulin gave better glucose control (mean below 7 mmol/L in 95.4% against 67.4%);
 - it went with larger infarct growth, a secondary outcome (median 27.9 against 10.8 cm³; P = 0.04);
 - three-month function was identical (45.6% in both groups);
 - deaths were 10% against 15.6%, not reported as significant.
- **Reading under the model:** correcting a visible figure (blood glucose) without adding resource reopens the insulin-dependent route into muscle and fat. Under the Selfish Brain account, post-stroke hyperglycaemia is the brain's own pull on supply (Sprengell, Kubera and Peters 2021b), so the deficit would move to the injured brain.
- **Why it is not support:**
 - one trial;
 - a secondary outcome at P = 0.04;
 - no difference in function or death;
 - where the glucose went was not measured;
 - the outcome was known before the model was applied.
- **What a test would need:** a pre-registered study in a system mapped as resource-short, measuring the delivery of the resource to each part before and after a correction that adds none, with the matching deficit predicted in advance.

S11. Dated record of the model's development

What this is. A dated record of the model's development, with the time stamps of each step. It is not evidence for the model.

- **Sources:** the public repository's commit history, and the author's earlier framework for institutions (dated 27 September 2026).
- **The earlier framework** is held privately and is available from the author on request. Its fingerprint (the SHA-256 of each file) is given in the public repository's README, so a copy supplied on request can be checked against it.
- **Wording:** "author" is the human author; "AI collaborator" is Claude (Anthropic).
- **Rule:** convergence with earlier work is reported as convergence, never as a successful prediction.

S11.1 How the model grew

Date (2026)	Step	What changed
By 27 September	The author's earlier framework for institutions	About 60 principles. Candidate formal models: a load ledger, priority queueing and a drift threshold
30 September	First notes on natural systems	Coupling and drift; living systems tie parts' fate to the whole; cancer as a part optimising for itself
3 October	Turn to natural systems	Load conservation stated as a principle; cross-domain cases gathered
4 October	Conserved quantity attempt	Hierarchy of parts, lower parts buffering higher ones, two directions of load, recovery order, part roles (store, intake, working part)
5 October	First queueing form of the model	Parts as servers with stores; load passed down; the break as a store running out
5 October	Twelve natural surface checks, each with predictions committed before sources were opened	Blood loss, fasting, kidney, honeybee colony, plants in drought, fetal growth restriction, rival recovery-order rules, muscle, the record rule, re-tuning, protection referenced to peak load, weight cycling
5 October	Priority as a formula	Order derived from marginal value, not assigned
5 October	Outside reviews absorbed, each logged as a source	Network rather than tree; use in reverse; economising
6 October	Simplifications by the author	Work is local and load is displaced; repair as a store refilling; economising as the governor's action
6 October	The present architecture	The system dies, not its parts; a separate governor that only allocates; parts with no demand of their own; units as state; repair cut first; basal maintenance before support work
6 October	First held-out test (H1)	Pre-registered and frozen before any data were opened
7 October	Literature read; current version	Established mathematics imported (DEB forms, viability theory, max-flow min-cut); the formal instances in Section 2 found; propositions proved and checked

S11.2 Theorised before the matching work was read (class A)

How the work ran. The AI collaborator drafted. The author shaped the model mainly by contesting drafts that did not fit the author's principles or observations. Most class A ideas came from those corrections.

Idea (date fixed in the record)	Whose	Matching work, and when it was found	Agreement and difference
Load is conserved: relocated, not removed (earlier framework; restated 3 October)	Author	Bode integral and conservation of fragility (Csete and Doyle 2002; Doyle and Csete 2011), read 3 October, after the statement. Kleinrock's (1965) conservation law, named 5 October	Same form: priority moves cost but cannot remove it. Theirs holds across frequencies or within one queue; ours across the parts of a system (Proposition 1)

Idea (date fixed in the record)	Whose	Matching work, and when it was found	Agreement and difference
The governor is a separate function that only allocates; parts have no demand of their own (6 October)	Author, correcting a draft in which the governor set targets	Supply-chain form of the Selfish Brain (Peters and Langemann 2009) and the central governor (Noakes 2012), read 7 October	Same split between a ranking regulator and passive parts. Perceptual control theory (Powers 1973) was noted by the AI collaborator when the idea was recorded
Ordered draw: parts drawn in a fixed order from one flow (6 October)	AI collaborator, formalising the author's governor rule	Plant sink priority (Grossman and DeJong 1994; Minchin, Thorpe and Farrar 1993) and the hierarchy of ATP consumers (Buttgereit and Brand 1995), found 7 October	Same architecture in plants and in the cell; neither has unit loss, scars, a ledger or use in other domains
Repair is the first thing cut when supply falls (6 October)	Author	Triage theory (Ames 2006); Bobba-Alves, Juster and Picard (2022); translational arrest (Hochachka et al. 1996), found 7 October	Same order of cuts. DEB pays maintenance first; the model reconciles the two (basal upkeep first, renewal cut first)
Economising: the governor lowers demand and capacity together, on purpose and reversibly (6 October)	Joint	Balanced metabolic suppression (Hochachka et al. 1996), found 7 October	Same reversible, regulated reduction
A tumour acts as a cut in total supply, with the order among the other parts kept (7 October)	Author	DEB tumour-in-host models, found in a screening completed the same afternoon	Same treatment of a tumour as a competing sink. The author had not seen these records
With a store carrying the gap and its release not binding, the break comes at the same cumulative shortfall whatever the rate (G3, first clause, 5 October)	AI collaborator	Fast and slow haemorrhage in sheep (Scully et al. 2016); fast and slow drought in trees (Dai, Wang and Wan 2018), matched after	Matches the first clause. The AI collaborator knew both in outline beforehand

S11.3 Known when the idea was formed (class B: precedent, no independence claimed)

Idea	Source, and when it was known
Brain spared while organs lose mass in starvation	The Selfish Brain account and Krieger's organ data, logged by the AI collaborator on 3 October. The author did not read it; no claim is made either way
A hierarchy of parts recruited under stress	Miller's Living Systems Theory, read 4 October
Three states of a part (coping, stressed, compromised)	Selye's general adaptation syndrome, read 4 October (secondary source)
Saturation and sudden decompensation	Woods (2018), read 3 October
Renewal classes of tissues	Bizzozzero (textbook pathology)
The point at which a store begins to release	Cherel and Groscolas's fasting phases, read 4 October
The compensatory reserve in blood loss	Convertino's work, known when the blood-loss predictions were written on 5 October
Shared upstream supply and a threshold switch	Built from blood-loss findings after the first natural check, 5 October

S11.4 Imported (class C: cited as the basis)

Taken from existing work and cited as such:

- DEB forms (shrinking, synthesising units, proportional release);
- viability theory;
- max-flow min-cut;
- the newsvendor critical fractile;
- metabolic control analysis;
- Michaelis-Menten competition;
- greedy allocation by marginal value.

Elements adopted from outside reviews are logged in the repository with their source.

S11.5 The caveat that governs class A

- The AI collaborator helped build the model, and its training very likely included the Selfish Brain, DEB, allostasis, Hochachka's and Ames's work, and plant allocation models.
- Only the author's route is independent. The model's wording is not.
- Class A is therefore reported as ideas **theorised beforehand and later found in the literature**, not as predictions and not as evidence.
- The evidence that the architecture is real is the agreement among the four fields in Section 2, which developed separately.

Supplement references

References cited only in the supplement. Works cited in the main text are listed there and are not repeated here. Each entry was checked against Crossref, PubMed or OpenAlex. How much of each was read is stated where it is used.

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